

Project and Professionalism (6CS007)

PawConnect: A Web-Based Online Pet Adoption & Management System

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Abstract

PawConnect is a web-based Online Pet Adoption and Management System developed to simplify and improve the pet adoption process by providing a centralized, user-friendly, and efficient digital platform for pet seekers, pet owners, and administrators. The system allows users to register and manage accounts, search and filter pets based on their preferences, send adoption requests, book appointments, communicate through a support chat system, and share reviews and feedback regarding their experiences. The project was developed using modern technologies including .NET for backend development, Node.js for server-side operations, and MongoDB for database management to ensure scalability, flexibility, and efficient data handling. An incremental development methodology was followed, where each module such as user management, pet listings, adoption requests, appointments, reviews, and chat support was implemented and tested separately before full integration, helping improve software quality and maintainability. PawConnect aims to overcome the limitations of traditional pet adoption methods by creating a faster, more organized, secure, and accessible online environment that promotes responsible pet adoption and effective pet management while enhancing communication and interaction between users and pet owners.

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Introduction

1.1 Project Briefing

PawConnect is a web-based platform developed to simplify and improve the process of pet adoption, pet purchasing, and pet-related communication. The main purpose of the system is to connect pet owners, shelters, and people who want to adopt or buy pets through a single organized digital platform. Many people who want to adopt pets face difficulties in finding reliable information about available pets. At the same time, pet owners often struggle to find responsible adopters. PawConnect addresses this problem by providing a centralized system where users can easily search, communicate, and complete the adoption process.

The platform allows pet owners to create listings for pets that are available for adoption or sale. Each pet listing contains detailed information such as the pet's name, breed, age, gender, health condition, vaccination status, and images. This information helps potential adopters understand the pet better before making a decision. Users who are interested in adopting or purchasing a pet can browse through different listings and use search and filtering options to find pets based on their preferences such as breed, type, or age.

Once a user finds a pet they are interested in, they can send an adoption or purchase request to the pet owner through the system. The pet owner can review the request and decide whether to approve or reject it. This approval process helps ensure that pets are given to responsible individuals who are capable of providing proper care. The system also includes an appointment scheduling feature that allows adopters and owners to arrange a meeting so the adopter can interact with the pet before making a final decision.

An additional important feature of PawConnect is the chat system, which enables direct communication between users. The chat system allows potential adopters to ask questions about the pet, discuss care requirements, and clarify important details with the pet owner. This communication improves transparency and helps build trust between both parties. It also reduces misunderstandings and allows users to get quick responses without needing to leave the platform.

The problem domain of this project is the lack of a structured and reliable platform for pet adoption and pet-related communication. Currently, many pet adoptions happen through social media posts or informal networks, where information can be incomplete or unreliable. This can lead to confusion, unsuccessful adoption attempts, or even unsafe situations for pets.

PawConnect solves this problem by providing a secure and organized system that manages pet listings, adoption requests, user communication, and appointment scheduling in one place.

1.2 Aim

The aim of the PawConnect project is to develop a reliable and user-friendly digital platform that connects pet owners, shelters, and individuals who want to adopt or purchase pets. The system aims to simplify the pet adoption process, improve communication between users, and provide a structured environment where pets can find safe and caring homes. By organizing pet listings and adoption processes in a single platform, PawConnect aims to make pet adoption more efficient, transparent, and accessible.

1.3 Objectives

- To provide clear and detailed information about pets available for adoption or sale.
- To allow users to search and filter pets easily based on their preferences.
- To allow users to send adoption or purchase requests to pet owners.
- To allow pet owners to approve or reject requests.
- To allow users to schedule meetings with pets before adoption.
- To improve communication between users through features such as chat.
- To make the pet adoption process more organized and efficient.

2.1 Artefact

2.1.1 Functional Decomposition Diagram (FDD)

The Functional Decomposition Diagram (FDD) shows the hierarchical breakdown of the PawConnect system into smaller functional modules. It helps in understanding how the entire system is divided into different components that perform specific tasks.

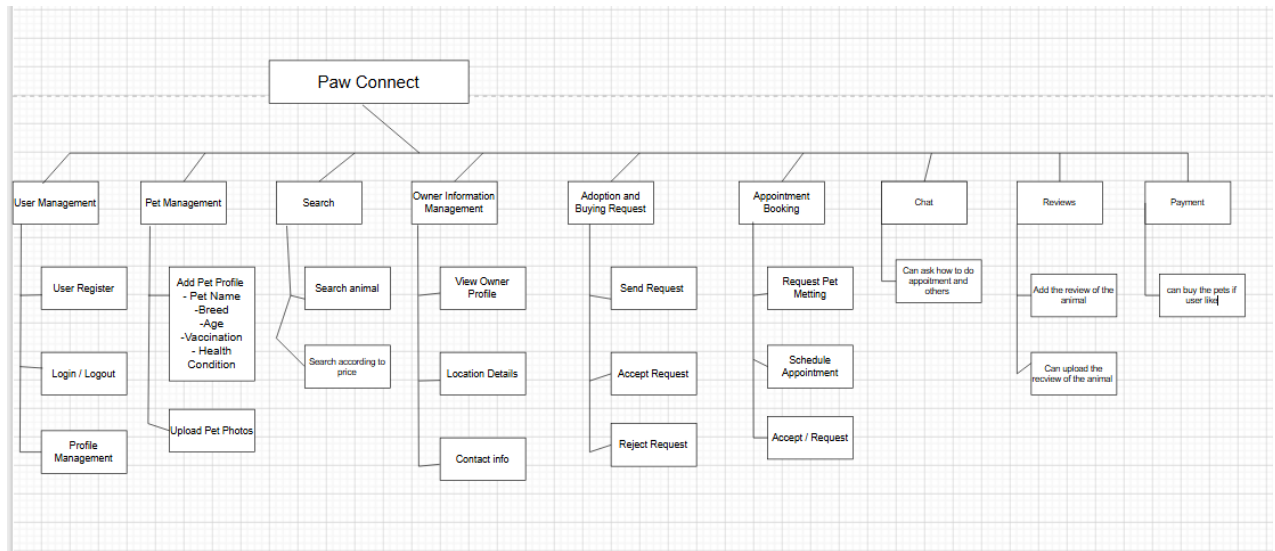


Figure 1: Functional Decomposition Diagram (FDD)

In the PawConnect platform, the main system is divided into seven major subsystems:

1. User Management
2. Pet Management
3. Search System
4. Owner Information Management
5. Adoption and Buying Request
6. Appointment Booking
7. Chat System
8. Reviews
9. Payment System

Each subsystem contains several sub-functions that collectively support the complete operation of the pet adoption platform. The decomposition of the system into smaller modules helps simplify development, testing, and maintenance.

2.1.2 Explain System as a Whole

PawConnect is a web-based platform designed to connect pet owners with people who are interested in adopting or buying pets. The system provides a centralized and user-friendly digital environment where users can search for pets, communicate with owners, and schedule meetings efficiently. The platform includes several integrated modules such as User Management, which handles user registration, login/logout, and profile management; Pet Management, where owners can add pet profiles with details like breed, age, vaccination status, health condition, and upload pet photos; and a Search System that allows users to search pets according to animal type and price. The system also includes Owner Information Management, which provides owner profiles, location details, and contact information. Through the Adoption and Buying Request module, users can send requests to adopt or purchase pets, while owners can accept or reject those requests. If a request is accepted, the Appointment Booking module allows users to schedule meetings with pet owners. In addition, the Chat System enables direct communication between users and owners to discuss pets or appointments. The Reviews module allows users to add feedback and reviews about pets and services, helping build trust within the platform. The Payment module supports secure online transactions for pet purchases or adoption-related payments. Overall, PawConnect simplifies the traditional pet adoption and buying process by integrating pet listing, communication, request handling, appointment scheduling, reviews, and payment services into one complete online platform.

2.1.3 Explanation of Each Subsystem

1. User Management System

The User Management subsystem handles all operations related to user accounts. It allows users to register, log in, log out, and manage their profiles.

This module ensures secure authentication and maintains user information such as usernames, email addresses, and passwords. It also manages user sessions and profile updates.

2. Pet Management System

The Pet Management subsystem allows pet owners to create and manage pet listings. Owners can add detailed information about their pets including breed, age, color, health condition, and photographs.

The system also allows owners to update or delete pet listings. This subsystem ensures that all pet information is properly stored and displayed within the platform.

3. Search System

The Search subsystem helps users find pets easily by providing search and filtering functionality. Users can search for pets based on different criteria such as pet type, breed, age, location, and health status.

The system retrieves relevant results from the database and displays them in a structured format for users to browse.

4. Owner Information Management System

This subsystem manages information related to pet owners. It stores details such as owner name, contact information, and location.

Users can view owner profiles when they are interested in a pet listing. This helps improve trust and transparency between pet owners and potential adopters.

5. Adoption and Buying Request System

The Adoption and Buying Request subsystem manages the process of submitting and processing adoption or purchase requests. Users can send requests to pet owners through the system.

Pet owners can review these requests and decide whether to accept or reject them. The system also notifies users about the status of their requests.

6. Appointment Booking System

The Appointment Booking subsystem allows users to schedule meetings with pet owners to see the pet before final adoption or purchase.

Users can request an appointment date and time, and the pet owner can either accept or reject the request. Once approved, the appointment details are stored and notifications are sent to both parties.

7. Chat System

The Chat System in PawConnect enables direct communication between users and pet owners within the platform. It allows users to start conversations from the pet details page to ask questions about pets and discuss adoption or buying details before submitting formal requests. Pet owners can respond through the chat interface, creating a two-way communication channel. The system also stores chat history so that both users and owners can review previous messages when needed. This feature improves interaction between both parties, provides better information about pets, and makes the adoption process more efficient and user-friendly.

8. Reviews System

The Reviews subsystem allows users to share feedback and experiences about pets, pet owners, and the overall adoption or buying process. After completing an adoption or purchase, users can add reviews and ratings through the platform.

These reviews help other users make informed decisions by providing information about the pet's condition, owner behavior, and service quality. The system stores and displays reviews publicly to improve transparency, trust, and reliability within the platform. It also encourages pet owners to maintain accurate pet information and provide better communication with adopters or buyers.

9. Payment System

The Payment subsystem manages all financial transactions within the platform. It allows users to securely make payments for pet purchases or adoption-related services through online payment methods.

The system processes payments, generates transaction records, and stores payment history for both users and pet owners. It also helps ensure secure and reliable transactions by protecting user payment information. This subsystem simplifies the payment process and provides a convenient way for users to complete pet buying or adoption procedures online.

2.2 Academic Question

How can a web-based information system improve the efficiency, communication, and management of the pet adoption and buying process?

2.2.1 Explanation of Academic Question

A web-based information system can improve the efficiency, communication, and management of the pet adoption and buying process by providing a centralized online platform where pet owners and adopters can interact easily. The system reduces the need for manual communication and physical visits by allowing users to search for pets, view pet details, and communicate with owners through digital features.

The platform improves efficiency through features such as online pet listings, search filters, request management, appointment booking, and secure user management. Users can quickly find pets based on their preferences, while pet owners can manage pet information and respond to adoption or buying requests more effectively.

The communication process is improved through the integrated chat system, which allows direct interaction between users and pet owners. In addition, appointment scheduling and notification features help coordinate meetings in an organized manner.

The system also improves management by securely storing user, pet, request, and payment information in a database. This makes the adoption and buying process more structured, transparent, reliable, and user-friendly compared to traditional manual methods.

2.3 Scope and Limitations of the Project

The PawConnect system is a web-based application developed to simplify and manage the process of pet adoption and pet buying. It provides a centralized platform where users and pet owners can interact efficiently and securely. The system aims to improve accessibility, communication, and transaction management within the pet adoption process.

The main scope of the project includes:

- User registration and account management for both adopters and pet owners
- Allowing pet owners to list pets available for adoption or sale
- Providing advanced search and filtering options to help users find suitable pets easily
- Enabling users to send adoption or purchase requests directly through the platform

- Allowing pet owners to accept or reject requests from users
- Providing appointment scheduling features for pet visits and meetings
- Integrating a review and rating system for users, pets, and overall experiences
- Including a chatbot assistant to help users navigate the system and resolve basic queries
- Supporting a payment gateway for secure online transactions during pet purchases

2.3.1 Limitations of the Project

Although the PawConnect system provides several useful features, it also has certain limitations that should be considered:

- The system is currently limited to a web-based platform and does not include a mobile application
- The chatbot feature may not handle complex or highly specific user queries effectively
- The review and rating system relies on user input, which may sometimes be biased or inaccurate
- Payment gateway integration depends on third-party services and may involve transaction fees or restrictions
- The platform does not include advanced verification of pet medical or veterinary records
- The system is designed for small to medium-scale usage and may require further optimization for large-scale deployment

2.4 Report Structure

This report is organized into several sections that describe the design, development, and evaluation of the PawConnect system.

The report begins with an introduction and abstract, which explain the purpose of the project, the problem it addresses, and the overall motivation behind developing the system. It then presents the Software Requirements Specification (SRS), which defines both the functional and non-functional requirements of the system in detail.

The next section focuses on the system design, where the architecture of the system is explained using UML diagrams. This includes use case diagrams, class diagrams, activity diagrams, sequence diagrams, and ER diagrams to represent different subsystems and their interactions.

Following the design section, the report includes wireframes and testing results, which demonstrate the user interface structure and validate the functionality of the system through different test cases.

Finally, the report concludes with a discussion of system artifacts, academic analysis, project scope, limitations, and overall findings, providing a complete understanding of the development process and outcomes of the PawConnect system.

3.1 Literature Review

3.1.1 Development of a Web-Based Pet Welfare Community (Mahadi & Alduais, 2024)

Summary: This paper presents a web-based pet welfare community designed to bring together adoption listings, lost-and-found pet notifications, and general community engagement in one place. The platform aims to centralise pet-related information so users can easily search, filter, and interact with data about pets in their area. It also encourages community participation by providing discussion forums and communication tools that connect pet owners, rescuers, and concerned citizens. The system successfully improves visibility and accessibility of welfare services but remains limited in terms of adoption workflow and data richness.

Pros: Centralises pet welfare information, improves discoverability, encourages community participation.

Cons: Focuses on listing and community forums; does not include rich pet metadata, owner-mediated adoption workflows, or integrated payment systems.

How PawConnect builds on it: PawConnect extends the platform's centralised approach by including detailed pet and owner profiles, structured adoption requests with owner approval, visit scheduling, and secure payment integration.

3.1.2 Shelter Soul: Bridging Shelters and Adopters Through Technology (Jagtap, 2025)

Summary: This study introduces Shelter Soul, a digital platform designed to modernise animal shelter operations by automating pet–adopter matching, managing donations, organising volunteer activities, and offering shelter analytics. The system focuses on improving internal efficiency for

shelters and enhancing the adopter experience through streamlined browsing and matching tools. Its architecture supports scalability and reliable shelter data management. However, because it is heavily shelter-oriented, it does not support private-owner pet listings or adoption workflows that involve direct owner approval.

Pros: Comprehensive shelter management; includes automated matching, donations, and analytics; modern scalable architecture.

Cons: Primarily shelter-centric; owner-mediated adoption requests and integrated payments are not included.

How PawConnect builds on it: PawConnect uses similar technical architecture and matching logic but adds private-owner adoption, and payment processing.

3.1.3 Pet Adoption System Using Web Technology (Bhadane et al., 2023)

Summary: This paper outlines a practical web-based adoption system that helps animal shelters manage their pet listings more efficiently. The platform allows pets to be categorised and searched by age, breed, and location, making it easier for adopters to find a suitable match. It improves data organisation for shelters by standardising how pet information is uploaded and displayed. Although it enhances basic adoption processes, the system includes only limited metadata about pets and lacks structured workflows such as owner-approved requests, scheduling visits, or integrated payment features.

Pros: Practical, implements searchable listings, improves data management for shelters.

Cons: No owner-mediated request workflow or integrated payment system; limited metadata (no health/vaccination details).

How PawConnect builds on it: PawConnect extends this system with detailed pet and owner profiles, request/approval workflow, and secure payment processing.

3.1.4 Design and Implementation of Intelligent Pet Adoption and Rescue Platform (Ye et al., 2025)

Summary: This research presents an intelligent platform that integrates rescue operations with adoption services using matching algorithms and data analytics. The system evaluates multiple

factors to recommend pets that fit adopters' preferences and enables shelters to track rescued animals and their medical care. It enhances adoption efficiency through intelligent decision support and

data-driven insights. Despite its advanced technology, the platform does not support private-owner adoption listings and does not incorporate features like structured payment processing owner-mediated approvals.

Pros: Intelligent matching improves adoption efficiency; includes analytics for shelters; combines adoption and rescue.

Cons: Focused on shelters; lacks private-owner adoption workflows and payment integration.

How PawConnect builds on it: PawConnect adds private-owner adoption requests, payment integration.

3.1.5 Predicting Adoption Speed of Pets from their Online Profiles

Summary: This study focuses on identifying which characteristics in a pet's online profile influence how quickly it gets adopted. Using large datasets, the researchers analyse structured metadata such as breed, size, colour, age, and health information to predict adoption speed. Their findings show that high-quality and detailed metadata boosts adoption likelihood and helps platforms optimise how pets are presented online. While the study is useful for understanding adoption patterns, it does not propose a complete adoption system or workflow.

Pros: Highlights the value of detailed metadata; supports evidence-based adoption optimization.

Cons: Adoption workflow, PawConnect adds private-owner adoption requests, payment system included.

How PawConnect builds on it: PawConnect captures rich pet and owner metadata, supports analytics, and integrates workflow and payment systems, making predictive insights actionable for real-world adoptions.

3.1.6 Pet Adoption System Using Web Technology (Bhadane et al., 2023)

Summary:

This paper presents a web platform built to standardise adoption processes and improve data handling for shelters. It focuses on making pet listings easy to access, offering search filters based on breed, age, and location so adopters can navigate more efficiently. The system creates a more organised digital environment for shelters to manage data. However, it does not provide features such as owner approval steps, scheduling tools, payment integration, or detailed metadata like health and vaccination information.

Pros: Practical implementation, improves shelter data management, provides searchable listings for adopters.

Cons: Does not include owner-mediated request/approval workflow, integrated payment processing, or detailed metadata such as vaccination and health conditions.

How PawConnect builds on it: PawConnect extends this system by incorporating owner profiles, a full request/approval workflow for adoption, visit scheduling, and secure payment integration, while also capturing detailed pet metadata including vaccination status and health condition.

3.2 Research and Investigation

Before developing the PawConnect system, several existing studies, platforms, and research works related to pet adoption systems, online marketplaces, and recommendation systems were reviewed. The purpose of this literature review is to understand how similar systems work, what technologies they use, and what limitations exist in current solutions. This research helps in identifying useful ideas and improving the design of the PawConnect system.

Many online platforms today allow people to adopt or purchase pets. These platforms usually provide features such as pet listings, search options, and communication between users. However, many of them still lack structured processes for adoption approval, proper communication tools, and intelligent recommendation systems. Because of these limitations, users often find it difficult to identify suitable pets quickly.

Research in online marketplace systems also shows the importance of structured information, user interaction, and recommendation features. Systems that provide detailed information and communication tools improve user experience and increase the success rate of transactions or adoptions. Based on this investigation, PawConnect was designed to include structured pet listings, request management, appointment scheduling, and a chat system to improve communication between pet owners and adopters.

3.3 Similar Systems

3.3.1. Sneha Care – Animal Rescue & Adoption (Nepal)

Main Features and Functionality:

Sneha Care focuses on the rescue, rehabilitation, and adoption of stray and abandoned animals. It also provides shelter management services, vaccination programs, and basic health care for animals. Adoption information is mainly shared through websites and social media platforms.

Limitations / Areas for Improvement:

The system does not provide a centralized or searchable adoption platform. There is no structured request or approval workflow for adoption processes. Additionally, pet information is limited, and there is no payment integration for adoption-related transactions.

How PawConnect Improves / Extends / Solves Limitations:

PawConnect addresses these issues by providing a centralized and searchable database of pets. It introduces a complete adoption workflow including request submission, approval, appointment scheduling, and payment processing. It also provides detailed pet metadata such as breed, age, vaccination status, health information, and owner details.

3.3.2. Animal Nepal

Main Features and Functionality:

Animal Nepal works to promote animal welfare and responsible pet ownership. It supports adoption programs, vaccination initiatives, and stray animal management. The organization also conducts community education and awareness campaigns.

Limitations / Areas for Improvement:

The platform does not offer a fully digital adoption management system. Information about pets and owners is limited, and there is no integrated system for scheduling or payment processing.

How PawConnect Improves / Extends / Solves Limitations:

PawConnect enhances these gaps by enabling both shelter and private-owner adoptions through a single platform. It provides a structured workflow that includes adoption requests, approvals, and scheduling. It also ensures better transparency through detailed pet and owner information.

3.3.3. Petfinder**Main Features and Functionality:**

Petfinder is an international pet adoption platform that provides a large database of adoptable pets. It offers advanced search and filtering options based on breed, age, size, and location. It also provides educational resources for adopters.

Limitations / Areas for Improvement:

The platform mainly focuses on shelters and does not strongly support private pet owners. It has limited adoption workflow features and does not include integrated payment functionality.

How PawConnect Improves / Extends / Solves Limitations:

PawConnect extends this model by supporting both shelter and private-owner listings. It introduces a complete adoption lifecycle with request handling, approval, scheduling, and payment integration. It is also localized for Nepal, making it more relevant to regional users.

3.3.4. SPCA Singapore**Main Features and Functionality:**

SPCA Singapore provides adoption services for cats, dogs, and small animals. It is also involved in rescue operations, sterilization programs, welfare initiatives, and public education campaigns.

Limitations / Areas for Improvement:

The platform has limited filtering and search capabilities. Communication is often handled through external channels such as email or phone. It does not support private-owner listings and lacks integrated scheduling and payment systems.

How PawConnect Improves / Extends / Solves Limitations:

PawConnect improves these limitations by offering a centralized search and filtering system. It also includes in-platform communication, structured approval workflows, scheduling features, and payment integration. Additionally, it supports private-owner listings alongside shelters.

3.3.5. Kathmandu Animal Treatment Centre (KAT Centre Nepal)**Main Features and Functionality:**

KAT Centre focuses on stray dog rescue, treatment, sterilization, and rehabilitation. It also runs adoption programs for recovered animals and promotes awareness campaigns on animal welfare. Adoption information is shared through its website and social media platforms.

Limitations / Areas for Improvement:

There is no centralized searchable database for adoptable pets. Adoption processes are handled manually through phone calls or social media. The system lacks a structured workflow, and pet information is often limited.

How PawConnect Improves / Extends / Solves Limitations:

PawConnect provides a fully structured digital platform with a searchable database of pets. It introduces an end-to-end adoption workflow including request submission, approval, scheduling, and payment. It also ensures standardized and detailed pet information, improving transparency and accessibility.

Main Features and Functionality:

Humane Society International is a global animal welfare organization involved in rescue operations, adoption programs, and community outreach initiatives. It works with partner shelters to support animal welfare and adoption activities.

Limitations / Areas for Improvement:

The organization does not provide a dedicated online adoption listing platform. There is no searchable database for pets, nor support for private-owner adoption systems. It also lacks integrated scheduling and payment workflows.

How PawConnect Improves / Extends / Solves Limitations:

PawConnect improves upon these limitations by offering a centralized platform for both shelter and private-owner listings. It provides a complete adoption workflow including

scheduling, approvals, and payment processing. It also ensures transparency through detailed and standardized pet and owner information.

4. Project Methodology

4.1 Justification for Choosing the Development Methodology

During the development of the PawConnect system, no formal software development methodology such as Agile or Waterfall was strictly followed. However, the project was developed in a structured and step-by-step manner based on understanding the requirements and continuously improving the system during implementation.

The development process was carried out in small parts, where each module such as user management, pet listing, search functionality, request handling, and scheduling was designed, implemented, and tested individually. This approach helped in identifying errors early and making necessary improvements during development.

This informal and iterative style of development was suitable for the project because the requirements were gradually refined while building the system. It also allowed flexibility to make changes whenever needed without following a rigid framework.

Although no specific methodology was officially adopted, the overall development process followed a practical incremental approach, which is commonly used in small-scale web application projects like PawConnect.

4.2 Suitability of the Methodology for the Proposed System

Although no formal software development methodology was strictly followed during the development of the PawConnect system, the development process was carried out in a

structured and iterative manner. This informal approach is suitable for the proposed system because it aligns well with the nature of web-based application development.

The PawConnect system consists of multiple independent modules such as user registration, pet listing, search and filtering, request management, appointment scheduling, chatbot support, review system, and payment integration. These modules were developed and tested individually, which made it easier to manage complexity and ensure proper functionality of each part.

This step-by-step development approach is suitable for the system because the requirements were refined during implementation. As the project progressed, improvements and adjustments were made based on testing results and better understanding of user needs. This flexibility is important in a system like PawConnect, where features such as communication, scheduling, and transactions may require continuous refinement.

Furthermore, an incremental approach helps in identifying errors early, improving system quality, and reducing development risk. It also allows easier modification and enhancement of features in future versions, such as adding mobile application support or improving chatbot intelligence.

Overall, the development approach used for PawConnect is suitable because it supports modular development, continuous improvement, and flexibility, which are essential for building an effective and scalable pet adoption platform.

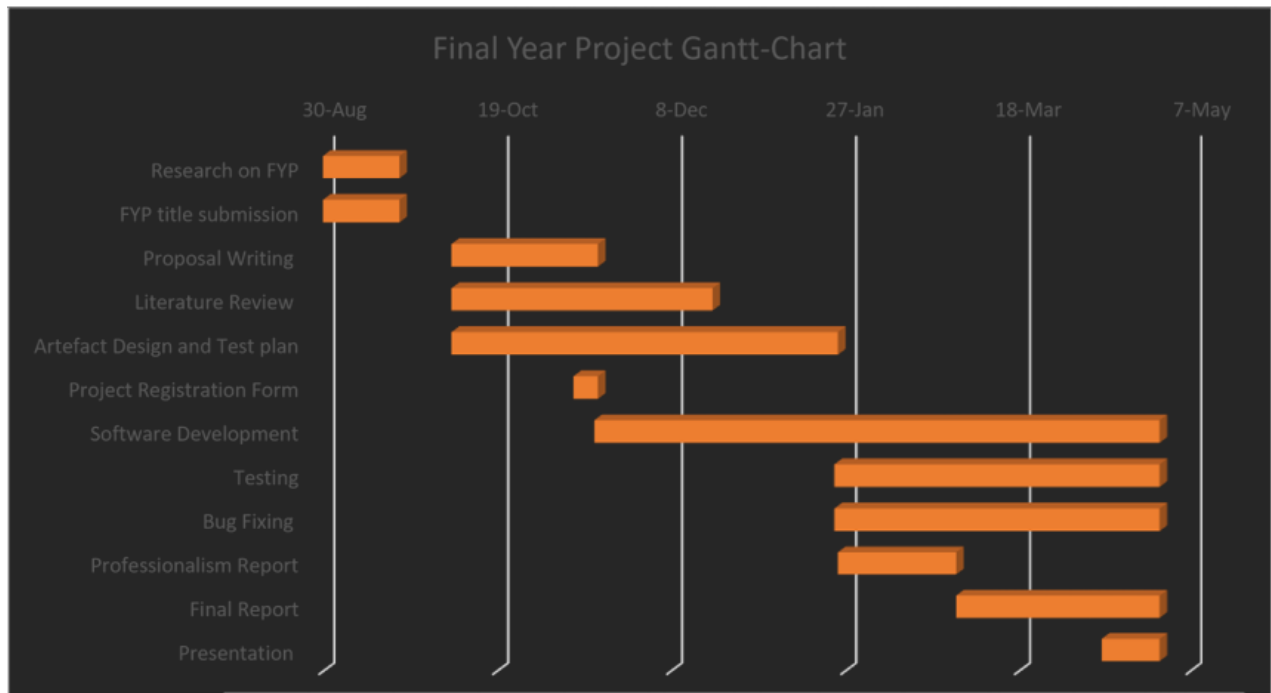


Figure II: Incremental Development Approach – PawConnect System

4.3 Project Milestones

The development of the PawConnect system was carried out through several major milestones. These milestones represent the key stages of the project lifecycle, starting from initial planning to final submission.

The main milestones of the project are as follows:

- Project proposal preparation and topic approval
- Literature review and background research on existing systems
- System requirement analysis, including identification of functional and non-functional requirements
- System design phase, including UML diagrams, database design, and overall system architecture
- System development and implementation of core features such as user management, pet listing, search, requests, and scheduling
- Testing and debugging to identify and fix errors in the system
- Documentation and report preparation, including system explanation and analysis

- Final system integration and submission

These milestones helped ensure that the project was completed in an organized and systematic manner, allowing proper tracking of progress throughout the development process.

5. Different Technologies and Tools Used for the Project

5.1 Justification for Selected Technologies

The PawConnect system was developed using modern web development technologies to ensure scalability, performance, security, and maintainability. Since the system involves user interaction, pet management, adoption requests, and real-time communication between users and pet owners, a full-stack web development approach was selected. The chosen technologies support a responsive user interface, efficient backend processing, secure data handling, and easy future enhancement of the system.

The following technologies and tools were used during the development of the PawConnect system:

5.1.1. Microsoft .NET

Microsoft .NET was used for backend development and server-side processing. It provided a secure and structured environment for handling application logic, API development, user authentication, and database connectivity. The framework helped in building a scalable and maintainable system architecture while improving overall application performance. Using .NET also simplified the management of business logic and backend services.

5.1.2. React

React was used to develop the frontend user interface of the PawConnect system. React enabled the creation of a responsive, interactive, and user-friendly web application. It helped in building reusable components, improving page rendering speed, and enhancing user experience. The technology also supported dynamic content updates without reloading the entire webpage.

5.1.3. MongoDB

MongoDB was used as the database management system for storing application data such as user information, pet details, adoption requests, and communication records. MongoDB is a NoSQL database that provides flexibility in managing large amounts of structured and unstructured data. It also supports scalability and efficient data retrieval, which is essential for modern web applications.

5.1.4. Figma

Figma was used for designing the user interface and creating system prototypes before implementation. It helped in planning the layout, navigation flow, and visual appearance of the application. Using Figma improved collaboration during the design process and allowed better visualization of the system before development started.

5.1.5. Postman

Postman was used for API testing and backend communication verification. It helped in testing HTTP requests such as GET, POST, PUT, and DELETE during the development process. Postman ensured that the APIs were functioning correctly and helped identify and debug errors efficiently.

5.2 Programming Languages and Frameworks

The PawConnect system uses React.js for front-end development and ASP.NET Core for backend development.

React.js was selected for the front-end because it allows the development of dynamic and responsive user interfaces. React uses a component-based architecture, which makes it easier to reuse UI components and maintain the system. It also improves performance by updating only the necessary parts of the interface when data changes.

For backend development, ASP.NET Core (.NET) was used. ASP.NET Core provides a robust and scalable environment for building web applications and APIs. It supports secure authentication, efficient data handling, and integration with databases. The framework also allows the development of RESTful APIs that can easily communicate with the React front-end.

This combination of React and ASP.NET Core provides a powerful full-stack architecture that supports both user interface functionality and server-side logic.

5.3 Development Tools

Several development tools were used during the implementation of the PawConnect system.

Visualstudio used as the primary Integrated Development Environments (IDEs). These tools provide features such as code highlighting, debugging tools, and extensions that improve development efficiency.

Git was used for version control to manage source code changes throughout the development process. It allows developers to track modifications, maintain different versions of the project, and collaborate more effectively.

5.4 User Interface Resources

To improve the visual design of the PawConnect system, several UI resources were used.

Font Awesome icons were used to provide graphical icons for features such as messaging, search, and navigation. These icons improve usability by helping users quickly recognize system functions.

Google Fonts were used to enhance readability and maintain a consistent and modern appearance across the user interface.

5.5 Testing Tools and Techniques

Testing was conducted to ensure that the PawConnect system functions correctly and reliably. Black-box testing was applied to test the overall functionality of the system without examining the internal source code. The testing process focused on verifying system inputs and outputs based on user requirements and expected behavior.

Different features such as user authentication, pet listing management, adoption requests, and chat functionality were tested to ensure they operated correctly from the user's perspective. The testing also verified whether the system responded properly to valid and invalid inputs, displayed appropriate error messages, and maintained smooth interaction between users and the system.

Black box testing helped identify functional errors, interface issues, and unexpected system behavior during development. It also ensured that all major components of the PawConnect system worked correctly before deployment, improving system reliability, usability, and overall performance.

5.6 Package Management Tools

The project was developed using React for the frontend and .NET for the backend. React was used to build an interactive and user-friendly interface, while .NET was used to develop and manage server-side functionalities.

6. Artefact Designs [in parallel with system development]

6.1 SRS (Software Requirement Specification)

- Functional requirements
- Non-functional requirements
- Constraints, assumptions, and dependencies

6.2 Design / Modelling Diagrams

6.2.1. User Management

1.1 Activity diagram

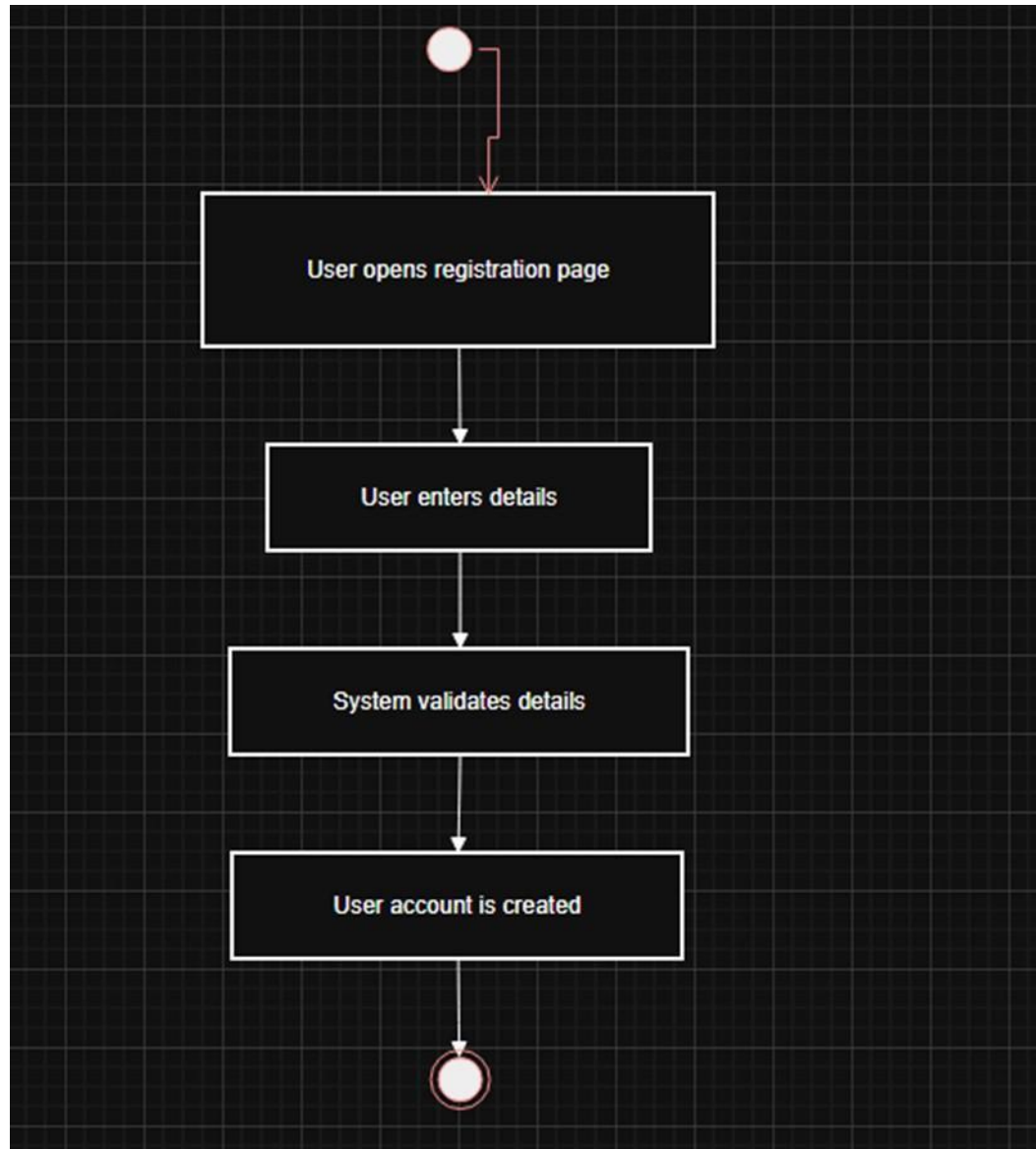


Figure III: User Management – Activity Diagram

This diagram is showing how users are registering, logging in, and managing their profiles. It is displaying the step-by-step flow from system entry to authentication completion and profile modifications.

1.2 Class Diagram

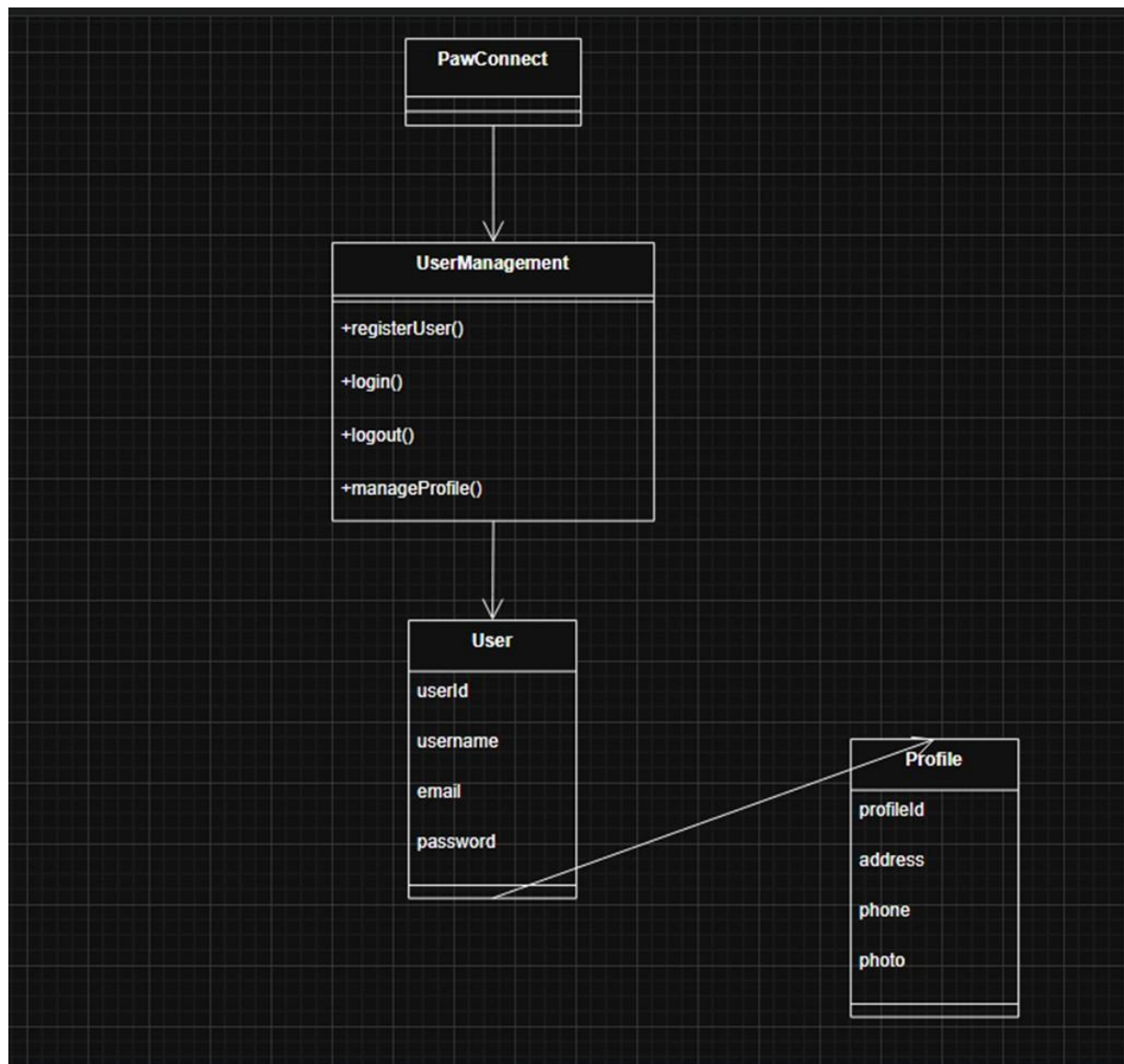


Figure IV: User Management – Class Diagram

This diagram is defining the User class and showing its attributes like userID, username, email, and password. It is also displaying methods that are handling registration, login, and profile updates.

1.3 Erd Diagram

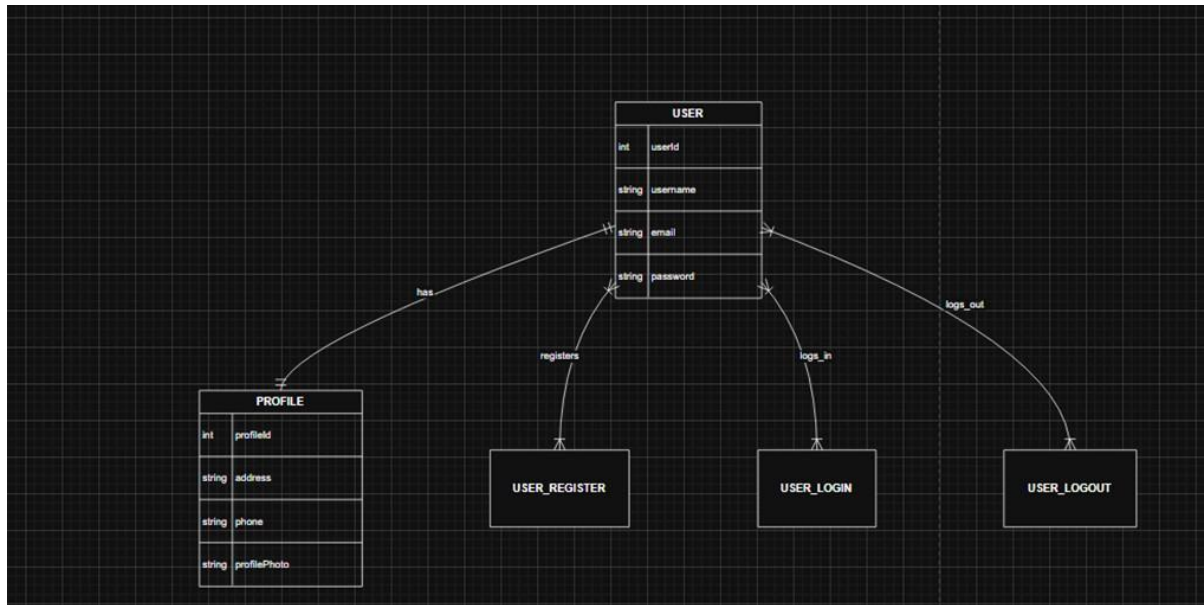
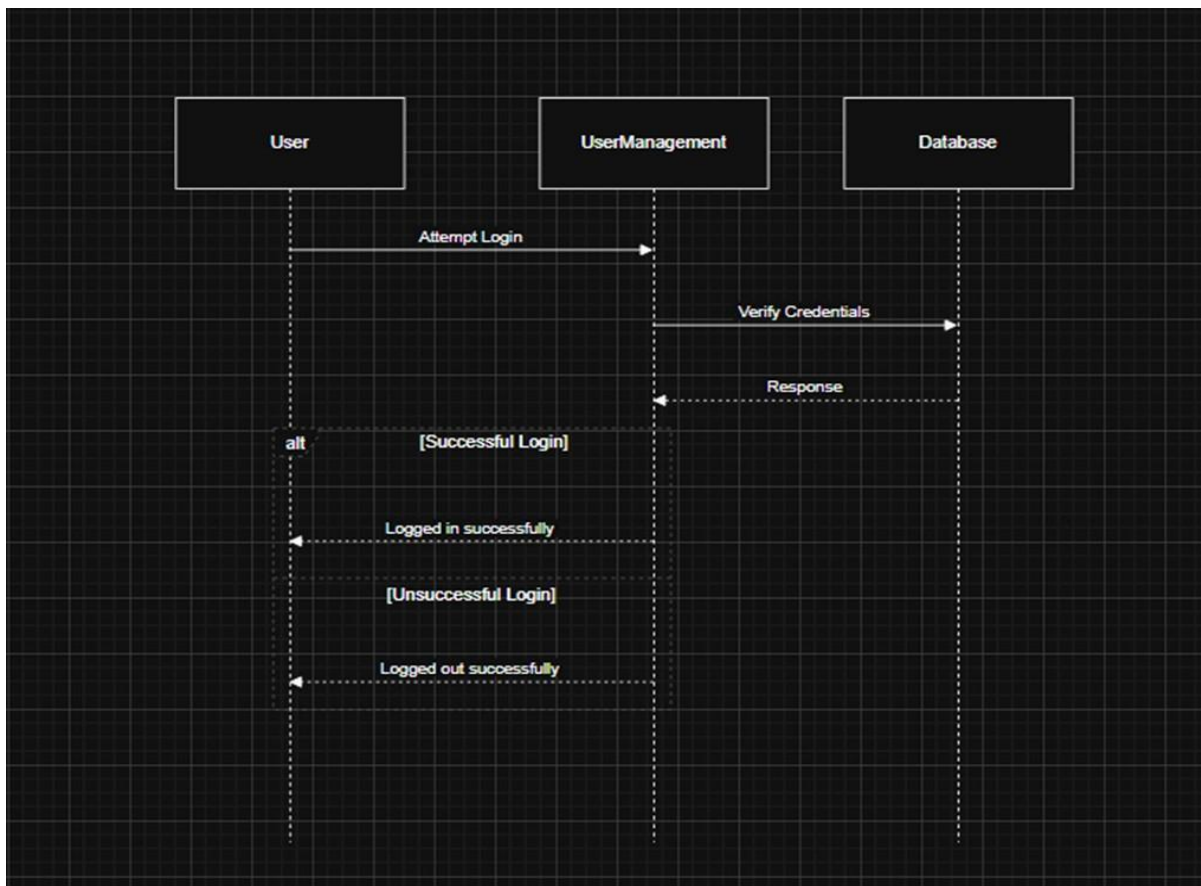


Figure V: User Management – ERD Diagram

This diagram is presenting the database structure for users and showing how the User entity is storing information and connecting with other tables in the system.

1.4 Sequence Diagram

Figure VI: User Management – Sequence Diagram



This diagram is demonstrating how the user interface, controller, and database are communicating during login and registration. It is showing messages flowing between these components in chronological order.

1.5 UseCase Diagram

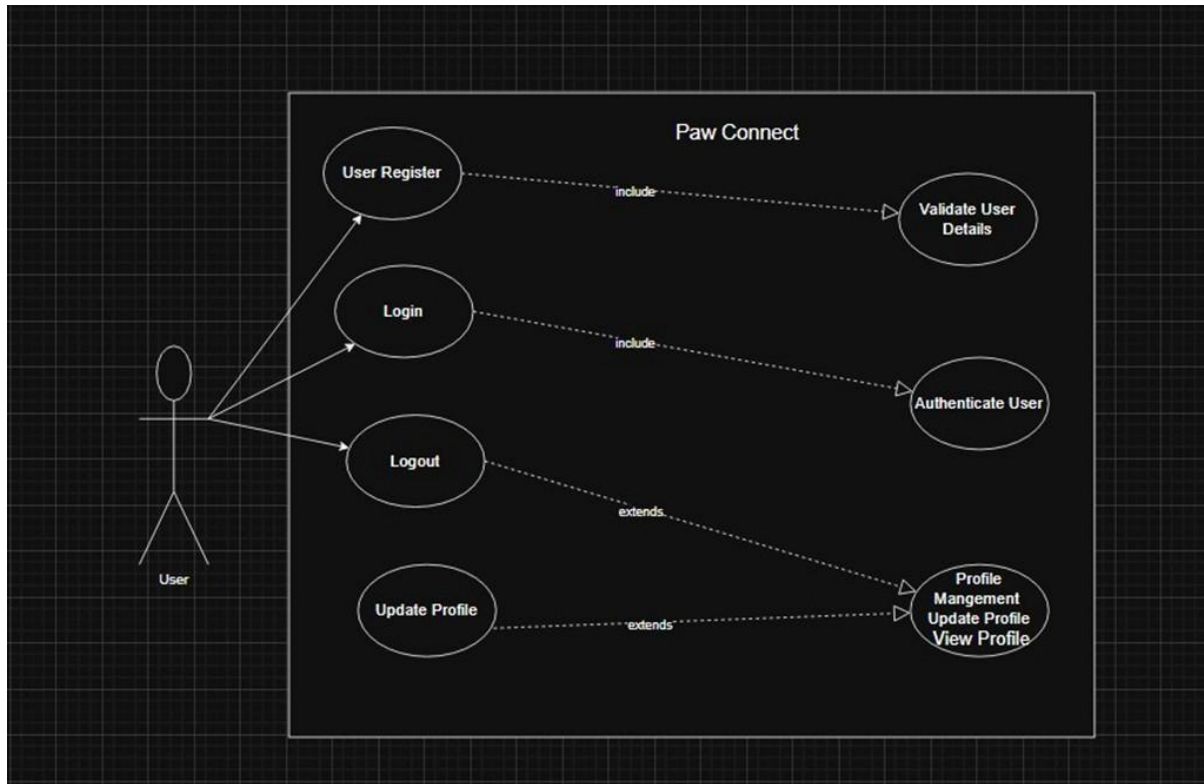


Figure VII: User Management – Use Case Diagram

This diagram is illustrating what actions users can perform and showing functionalities like registering, logging in, updating profiles, and logging out.

6.2.2. Pet Management

2.1 Activity Diagram



Figure VIII: Pet Management – Activity Diagram

This diagram is mapping out how pet information is being added, updated, viewed, and deleted. It is showing decision points and alternative paths in the workflow.

2.2 Class Diagram

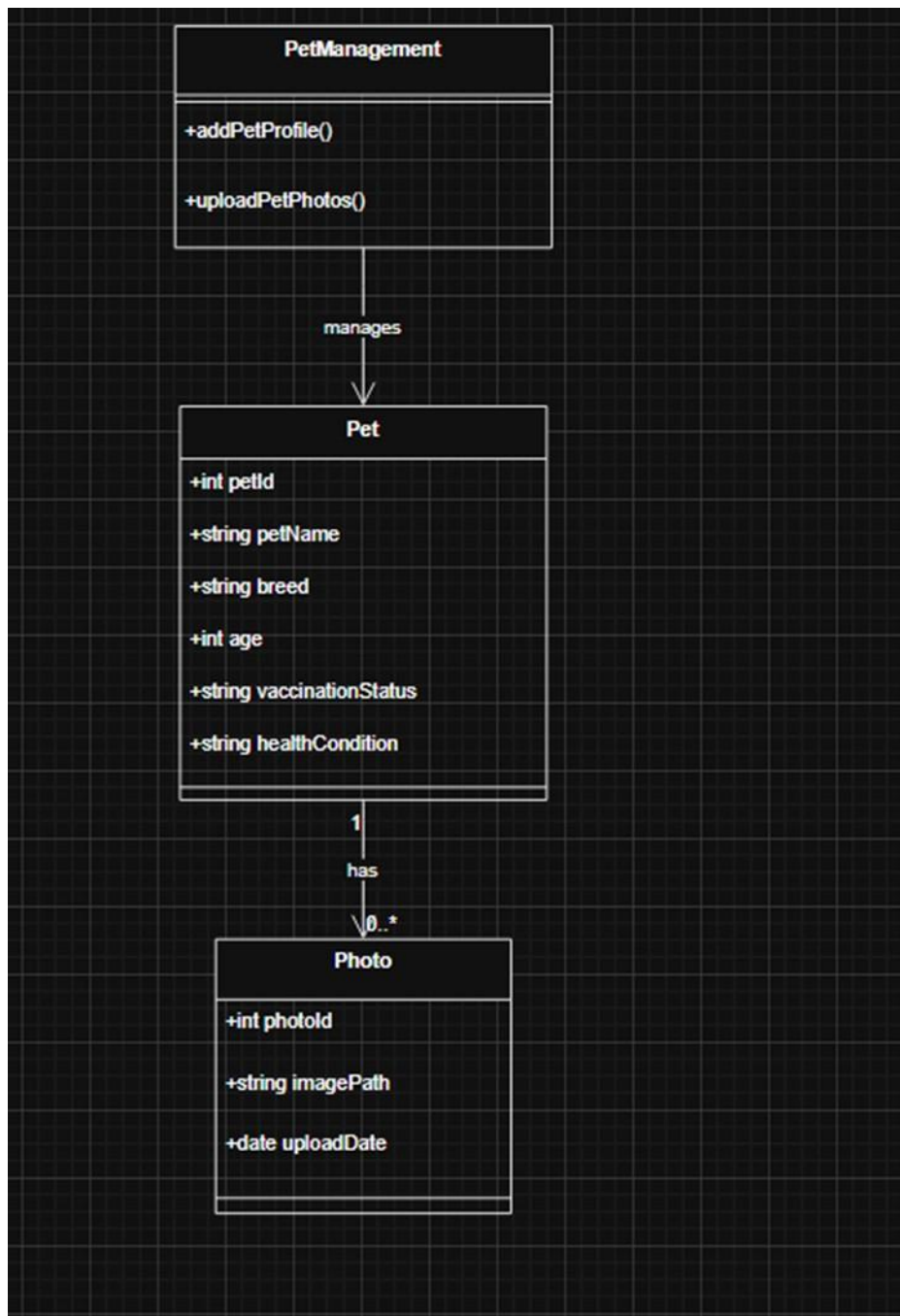


Figure IX: Pet Management – Class Diagram

This diagram is representing the Pet class and displaying attributes such as petID, name, breed, age, and health status. It is also showing methods that are performing create, read, update, and delete operations.

2.3 Erd Diagram

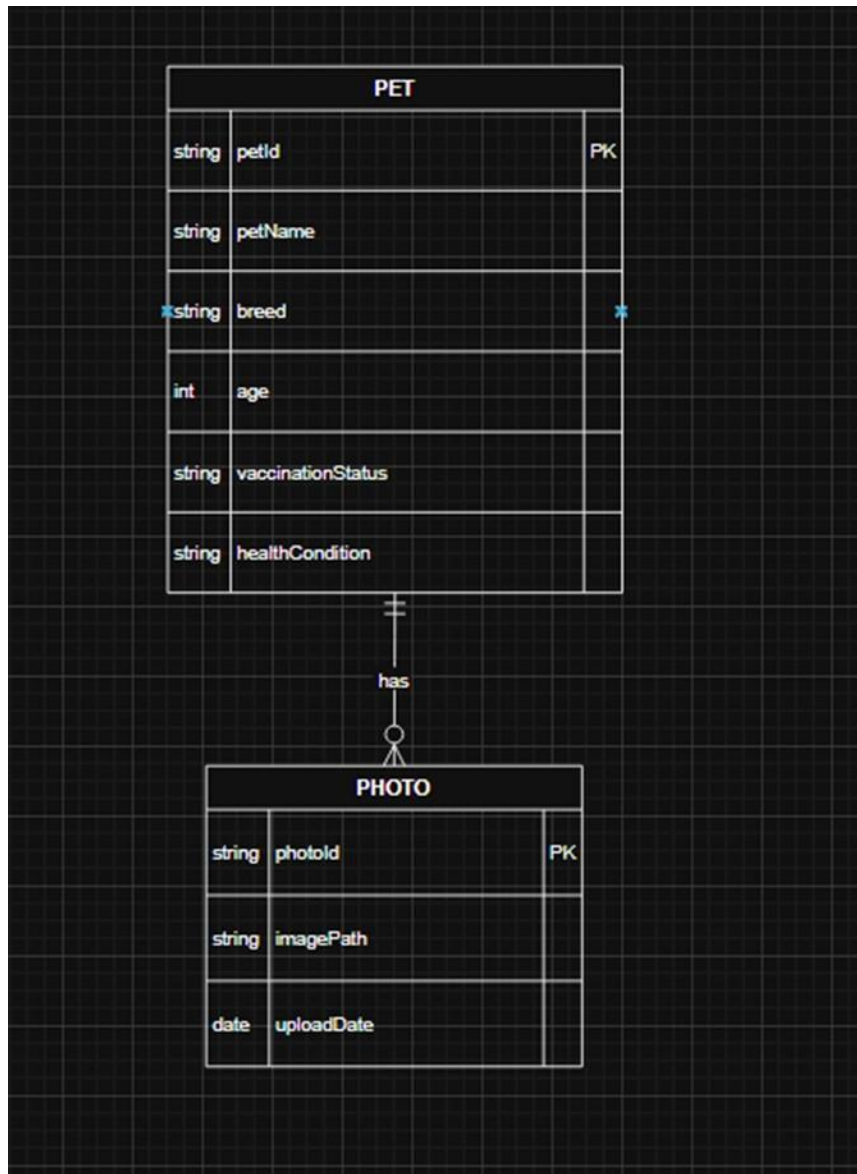


Figure X: Pet Management – ERD Diagram

This diagram is showing the Pet entity structure and illustrating how it is relating to User and Owner entities through foreign key relationships.

2.4 Sequence Diagram

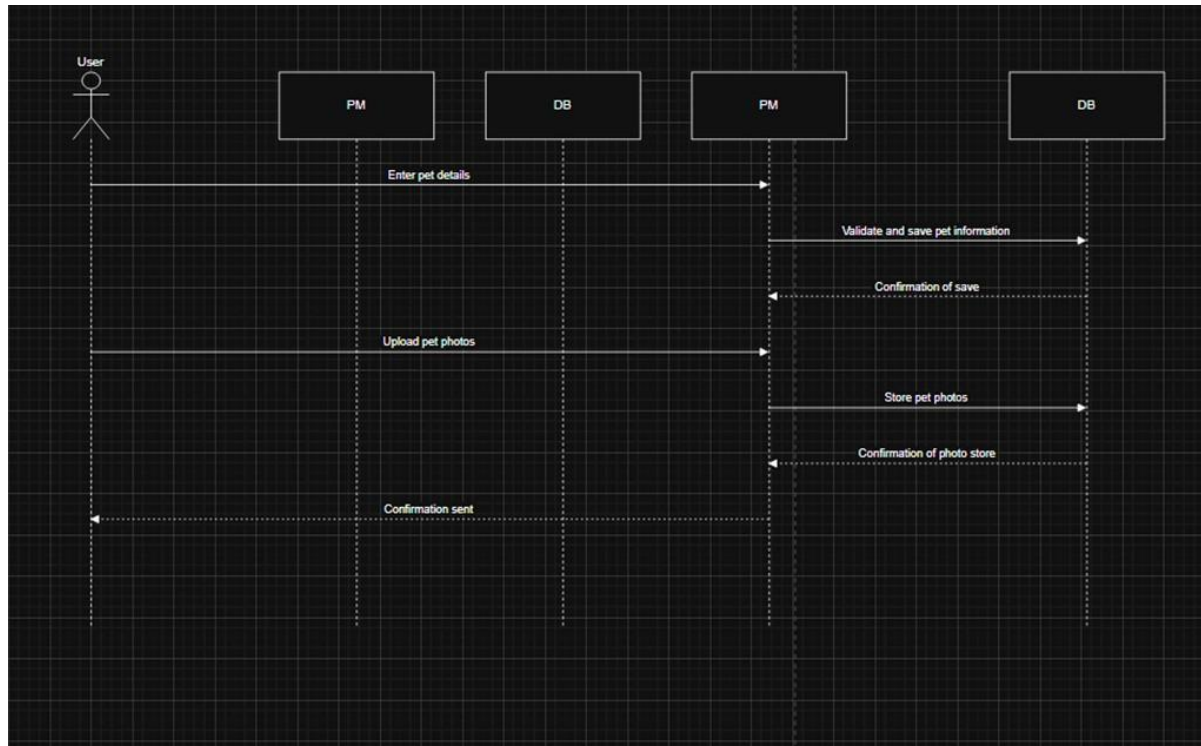


Figure XI: Pet Management – Sequence Diagram

This diagram is depicting how components are interacting when managing pet information. It is showing the flow from user requests through system processing to database operations.

2.5 UseCase Diagram

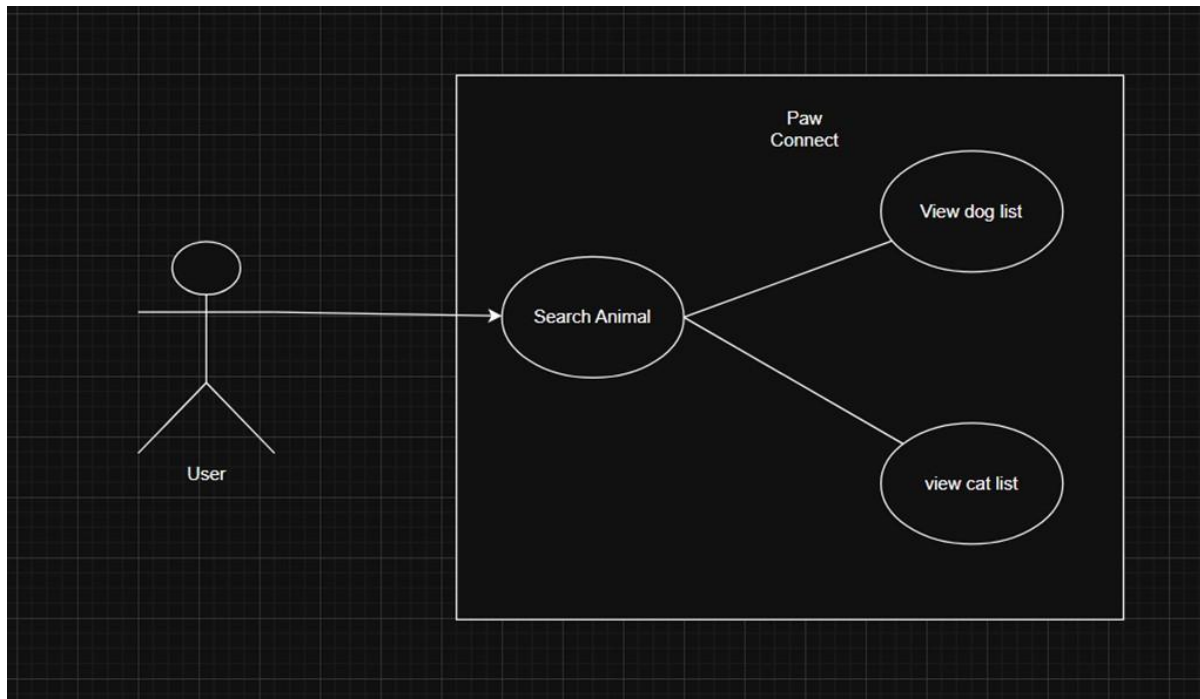


Figure XII: Pet Management – Use Case Diagram

This diagram is displaying what pet owners can do, showing actions like adding pets, viewing details, updating information, and removing pets.

6.2.3. Search

3.1 Activity Diagram

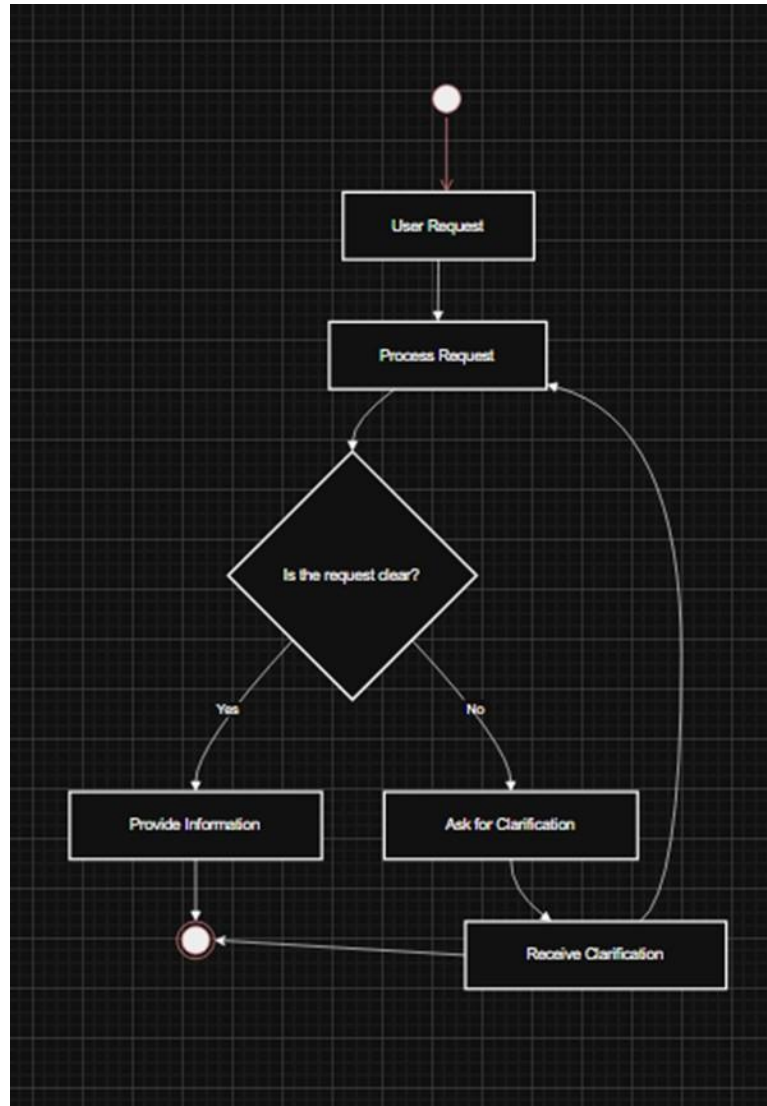


Figure XIII: Search – Activity Diagram

This diagram is illustrating how the search process is working, showing users entering criteria, filtering results, and viewing matching pets.

3.2 Class Diagram

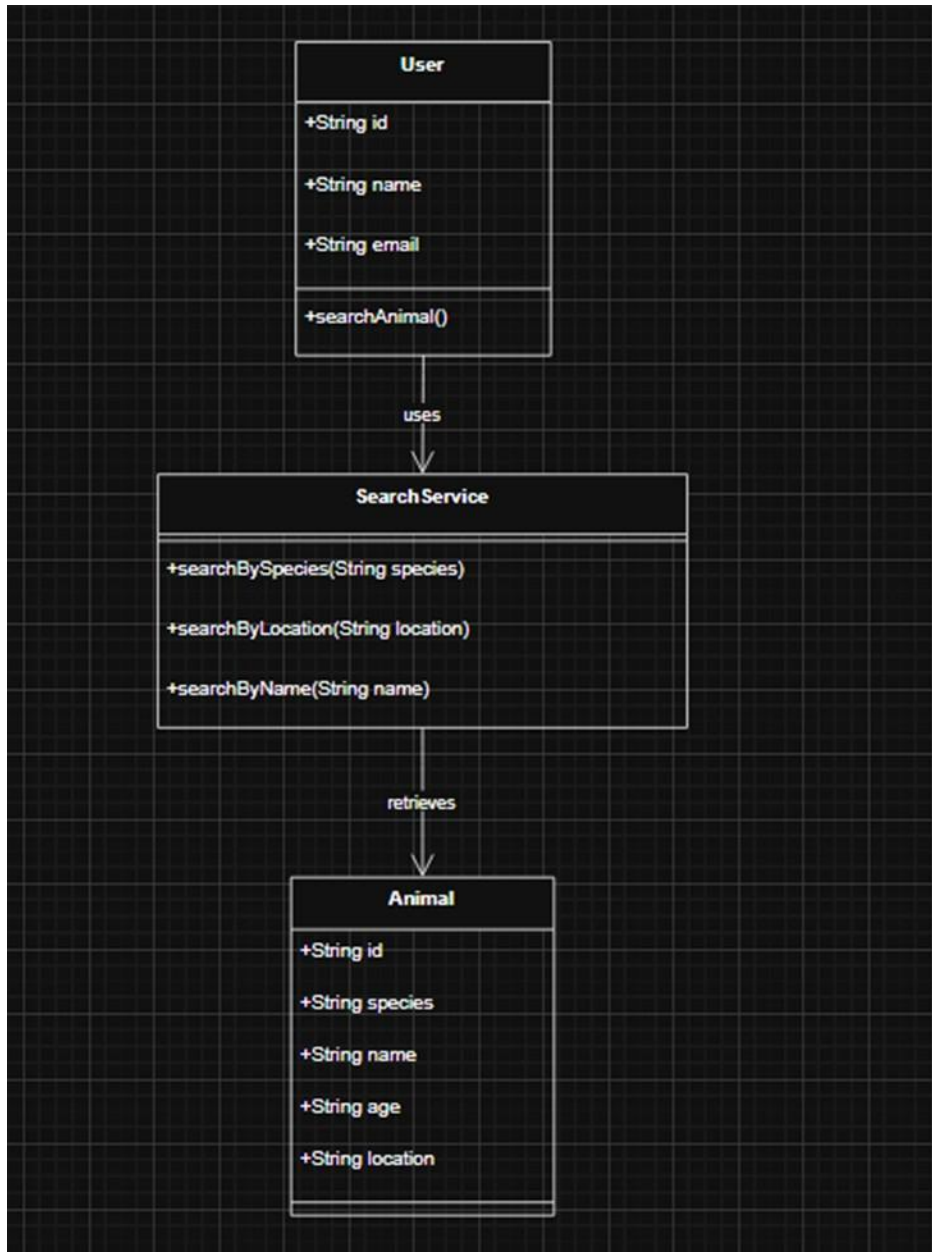


Figure XIV: Search – Class Diagram

This diagram is defining Search-related classes and showing attributes for search parameters like breed, age range, and location. It is displaying methods that are executing search queries.

3.3 Erd Diagram

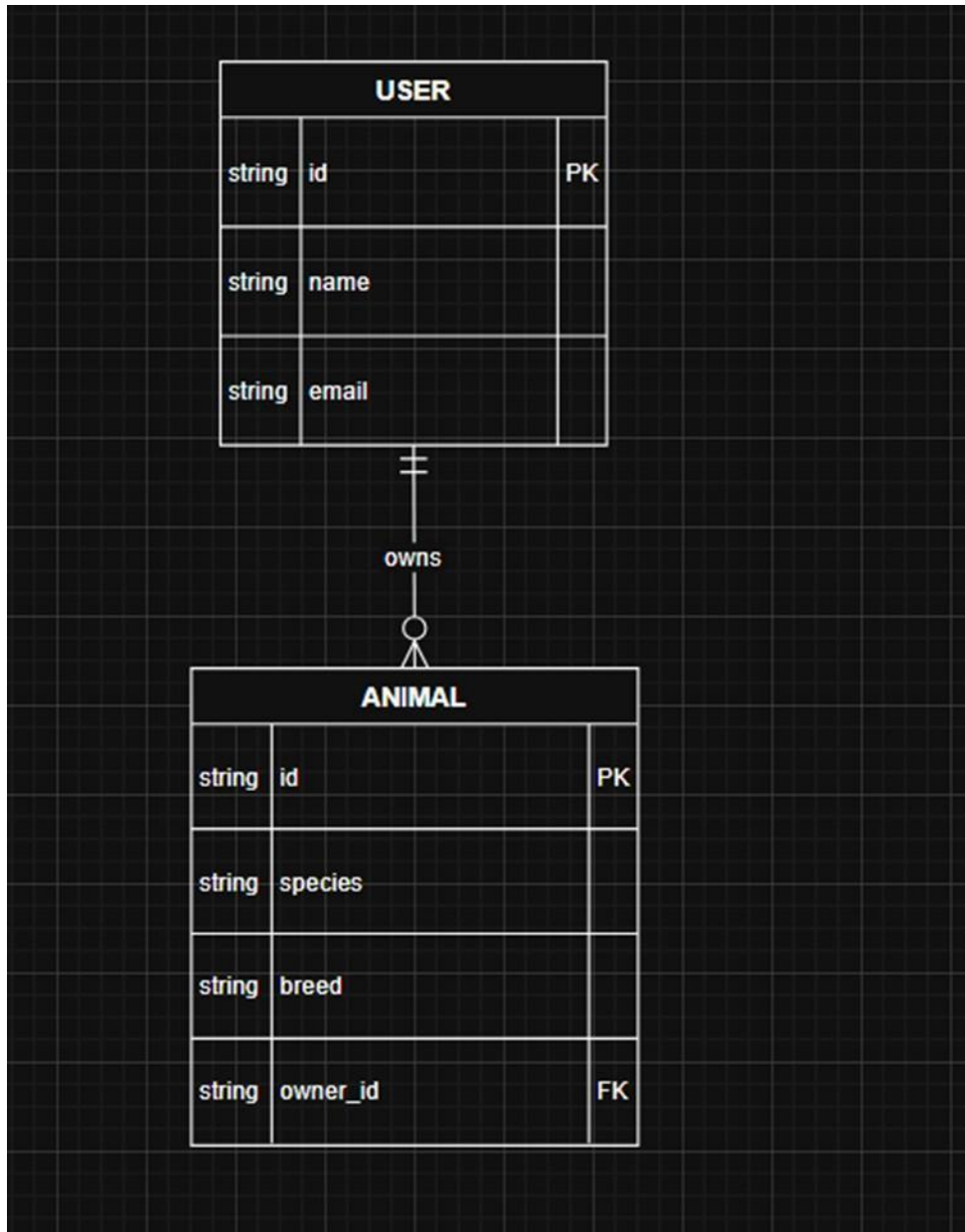


Figure XV: Search – ERD Diagram

This diagram is showing how search functionality is connecting to pet and user databases. It is illustrating indexing and filtering relationships.

3.4 Sequence Diagram

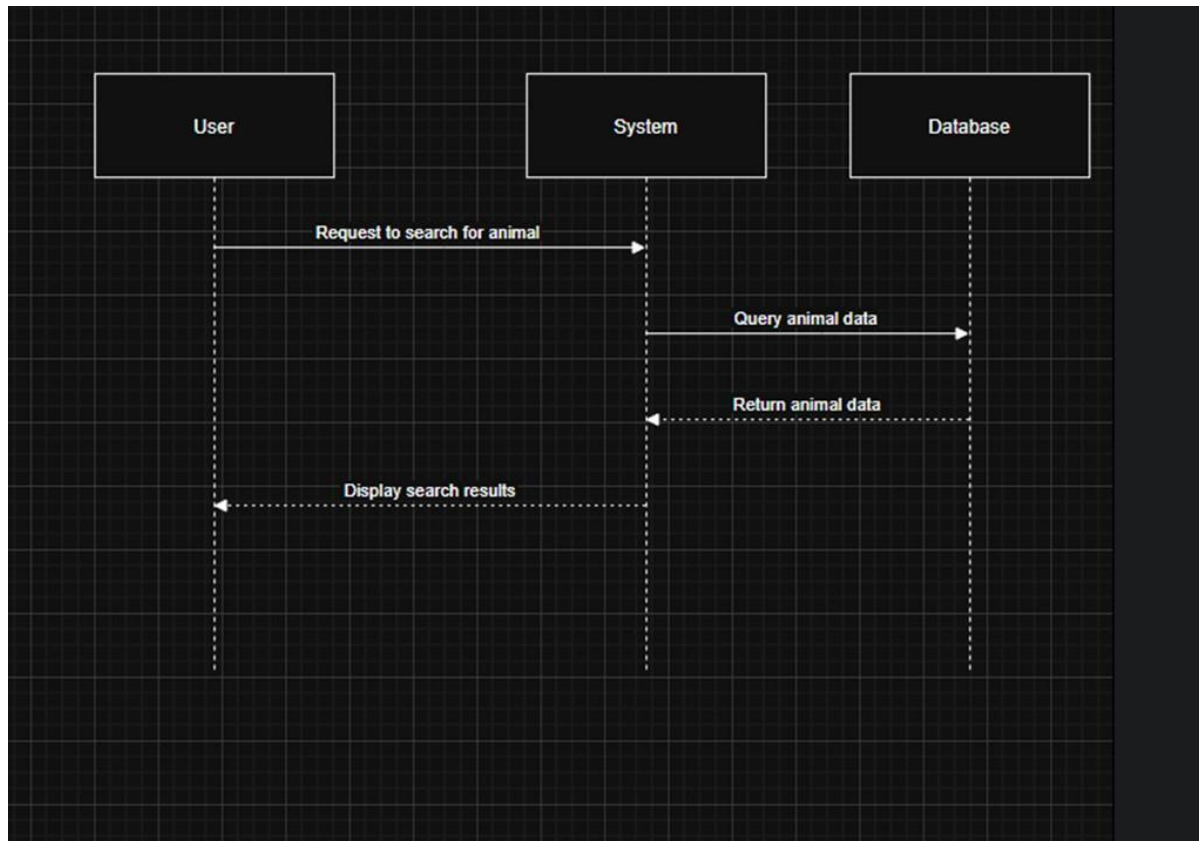


Figure XVI: Search – Sequence Diagram

This diagram is demonstrating how messages are exchanging between the search interface, controller, and database when users are performing searches.

3.5 UseCase Diagram

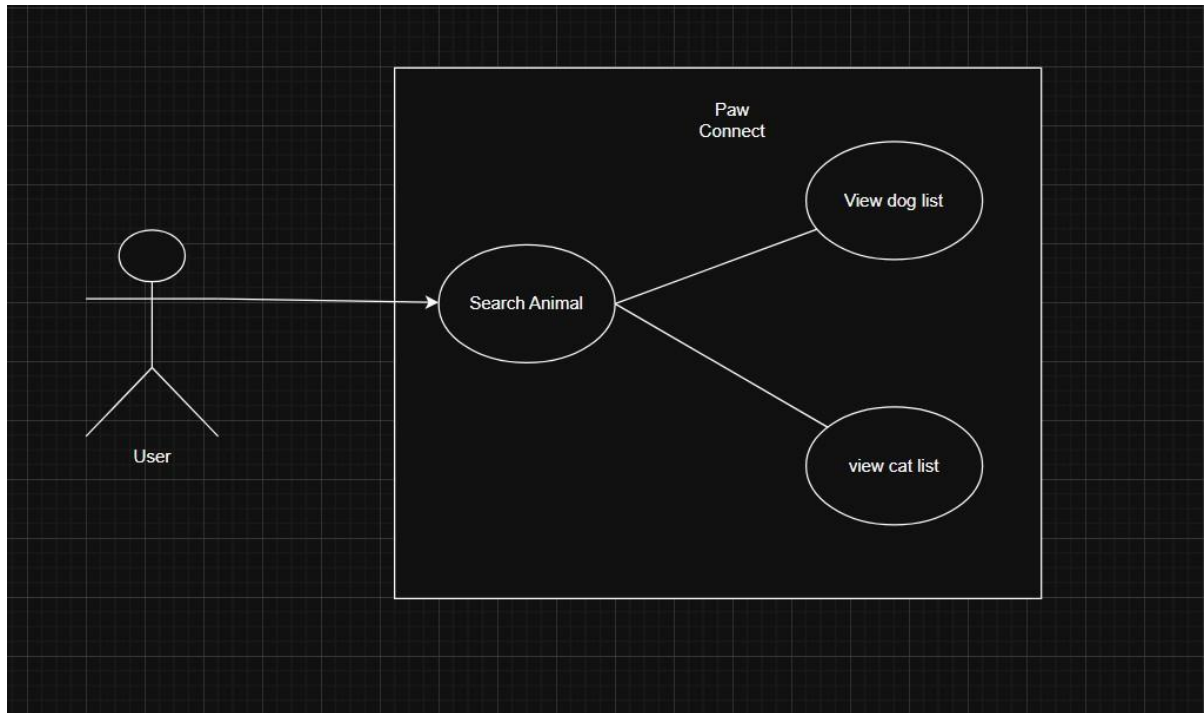


Figure XVII: Search – Use Case Diagram

This diagram is displaying search-related actions that users can perform, showing options like searching by breed, filtering by location, and viewing results.

6.2.4. Owner Information Management

4.1 Activity Diagram

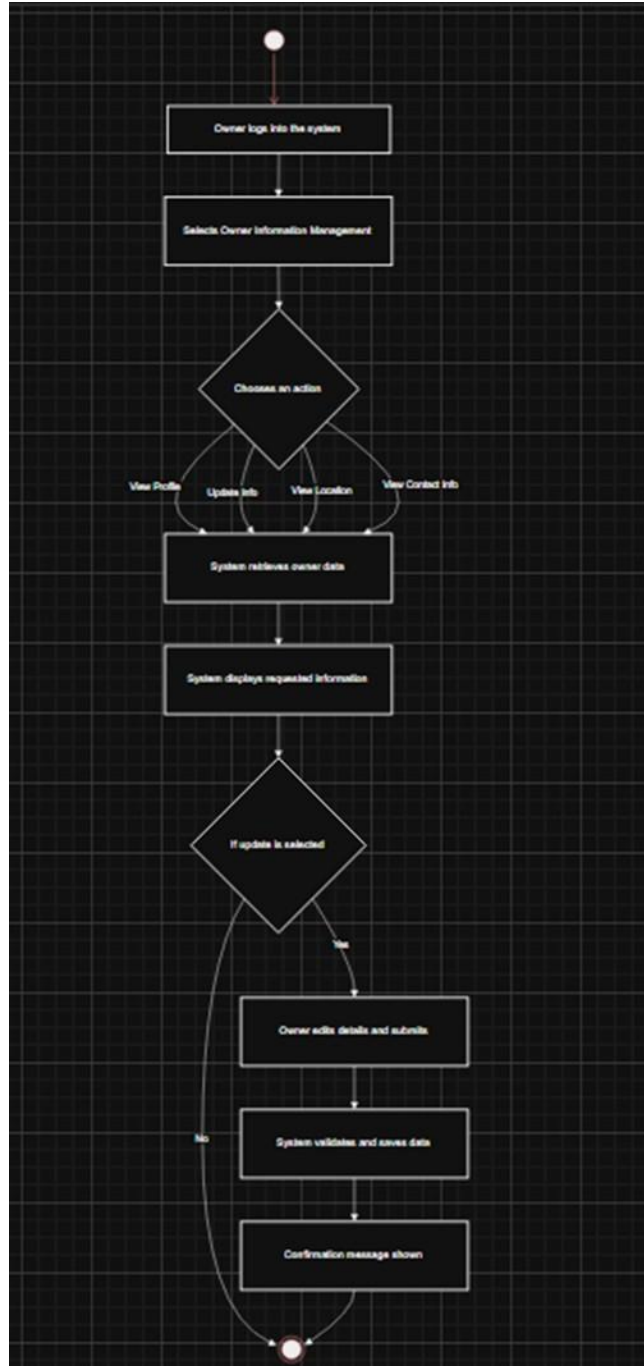


Figure XVIII: Owner Information Management – Activity Diagram

This diagram is showing how owner details are being managed, displaying the flow for adding, editing, and verifying owner information.

4.2 Class Diagram

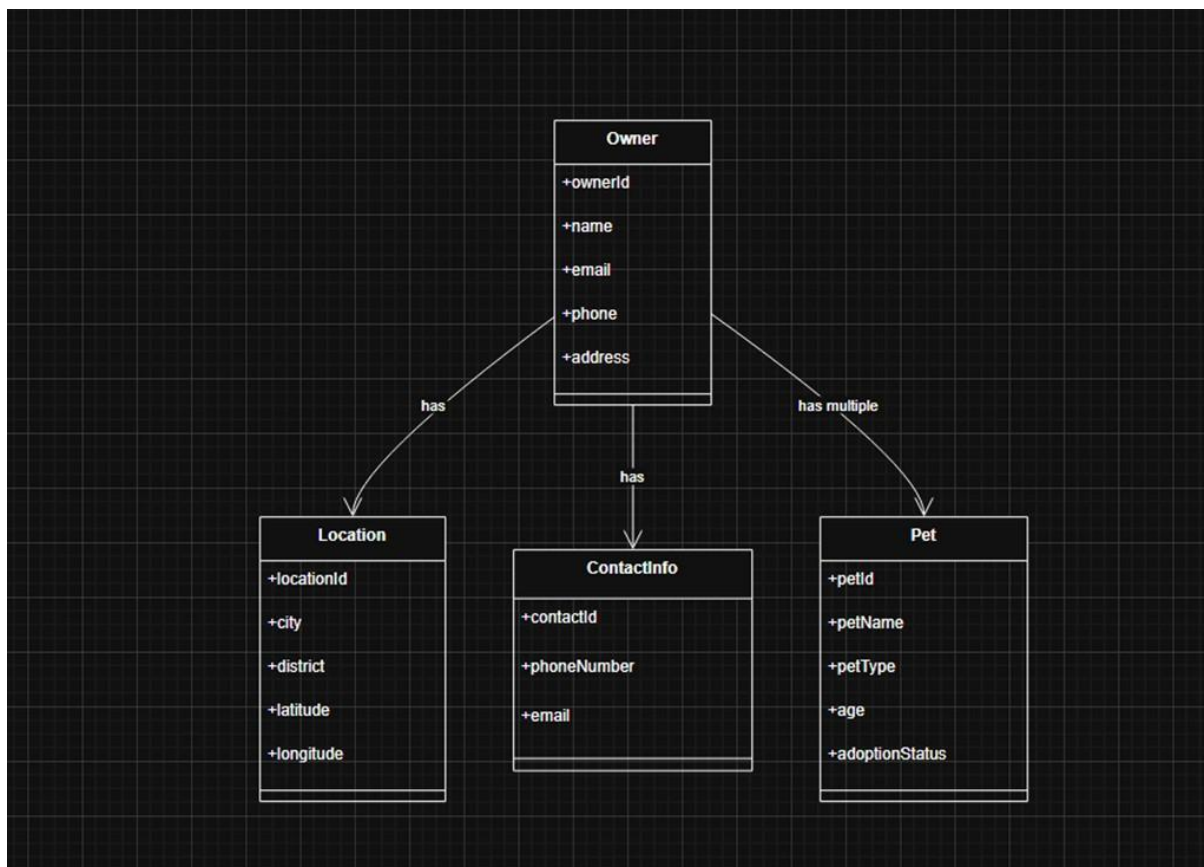


Figure XIX: Owner Information Management – Class Diagram

This diagram is representing the Owner class and showing attributes like ownerId, name, contact details, and address along with associated methods.

4.3 Erd Diagram

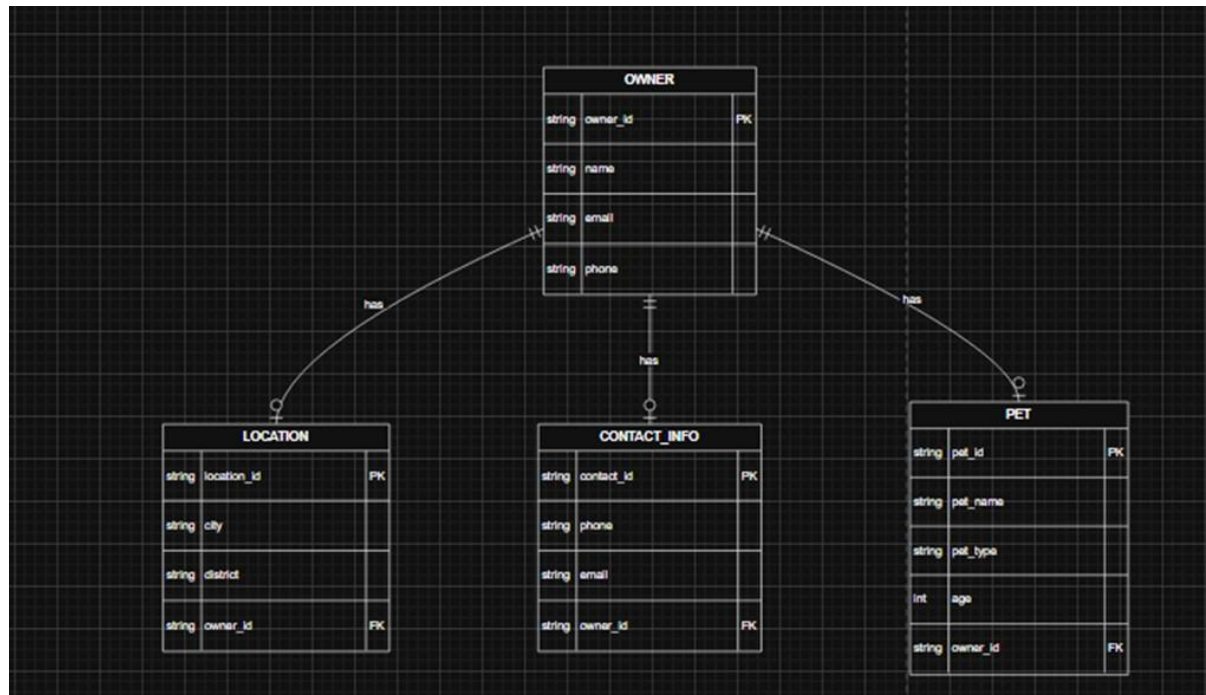


Figure XX: Owner Information Management – ERD Diagram

This diagram is displaying the Owner entity and showing how it is relating to pets and other entities through one-to-many connections.

4.4 Sequence Diagram

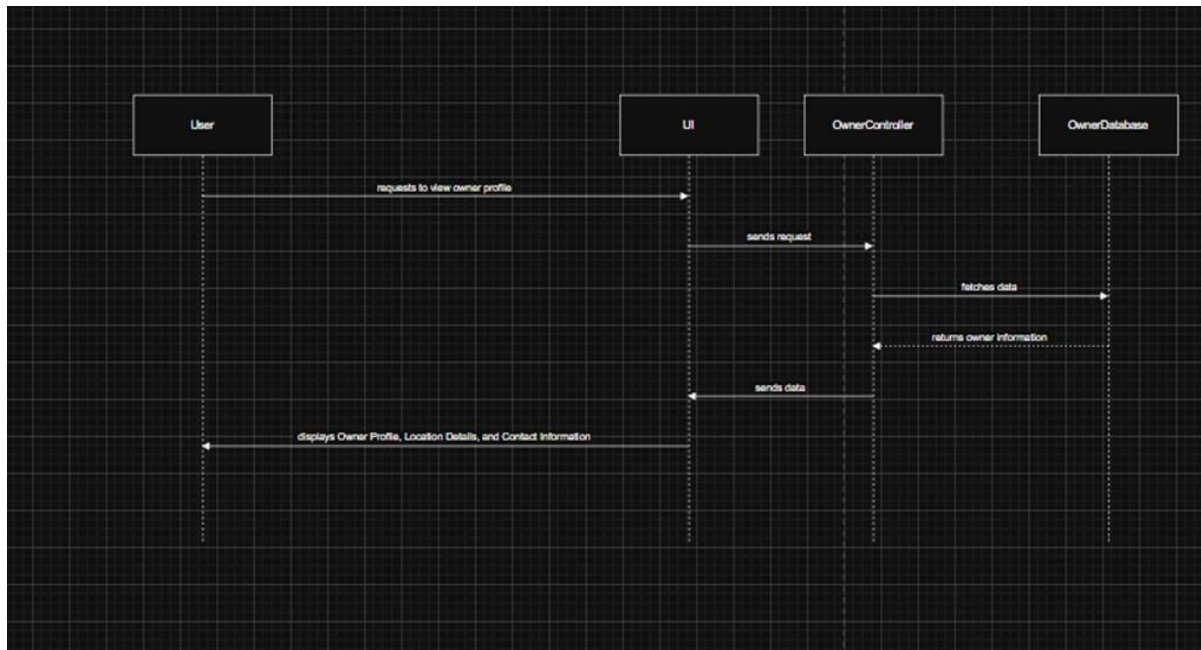


Figure XXI: Owner Information Management – Sequence Diagram

This diagram is illustrating how components are interacting when processing owner information updates and retrievals.

4.5 UseCase Diagram

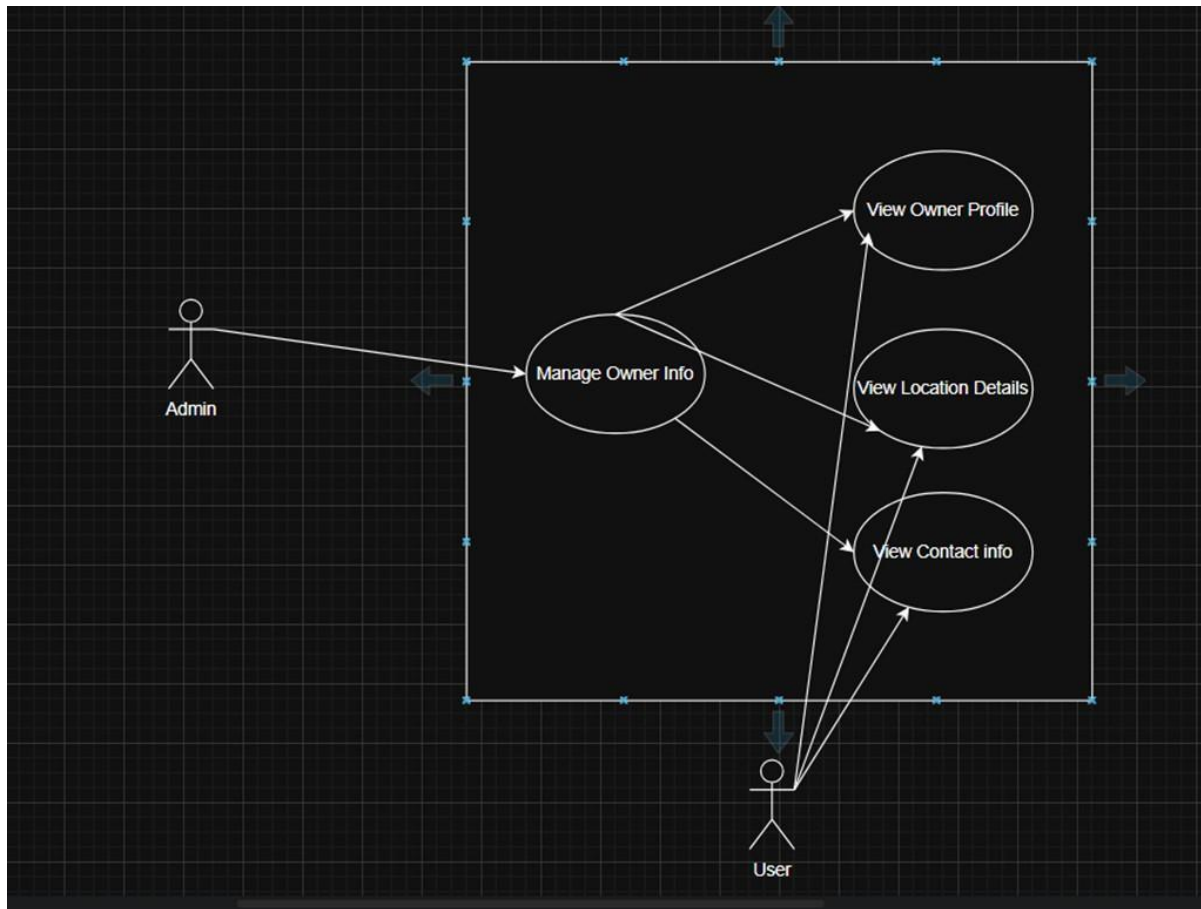


Figure XXII: Owner Information Management – Use Case Diagram

This diagram is showing what administrators and owners can do regarding owner information management.

6.2.5. Adoption and Buying Request

5.1 Activity Diagram

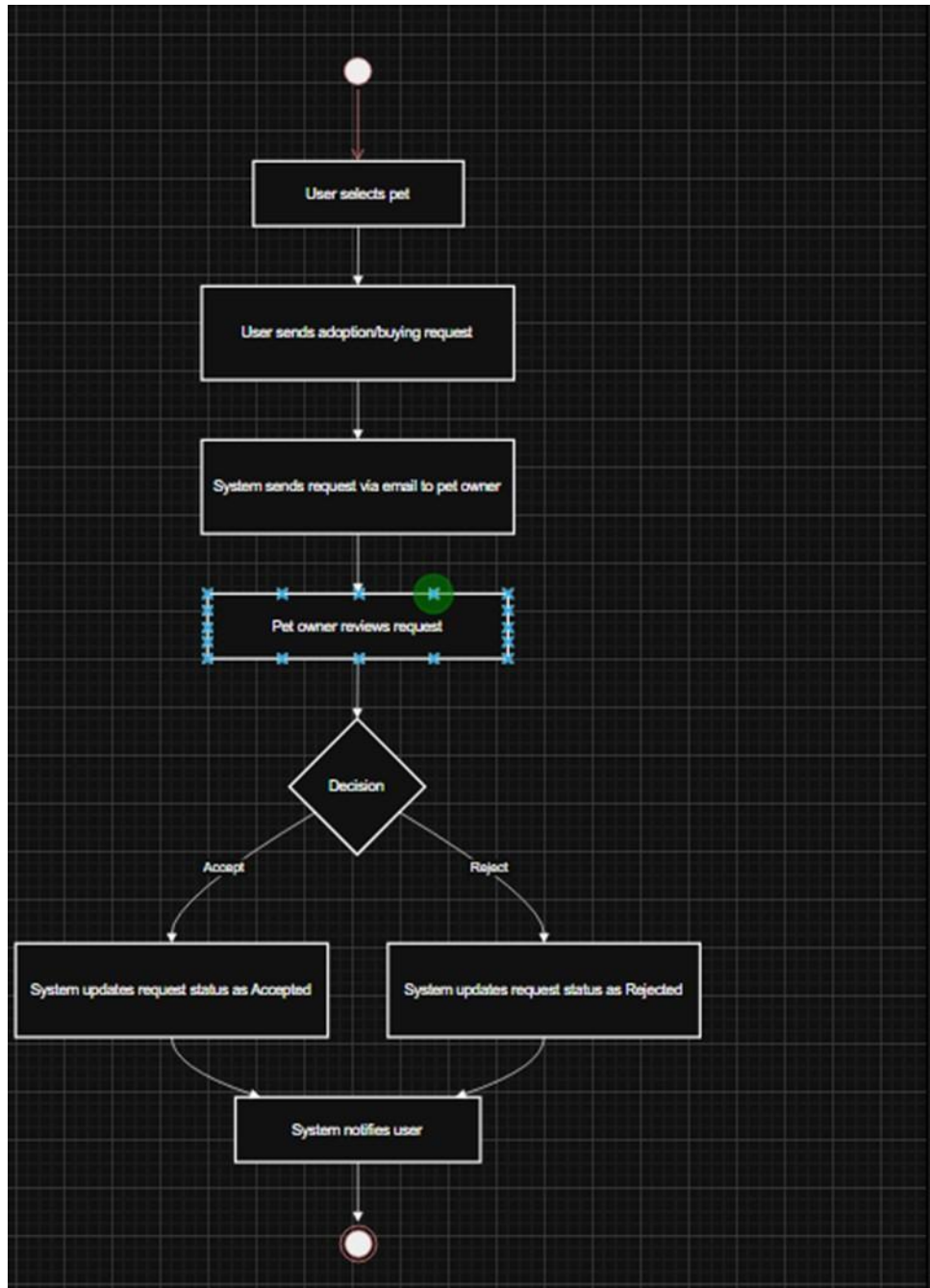


Figure XXIII: Adoption and Buying Request – Activity Diagram

This diagram is mapping the complete process from browsing available pets to submitting requests and going through approval workflows.

5.2 Class Diagram

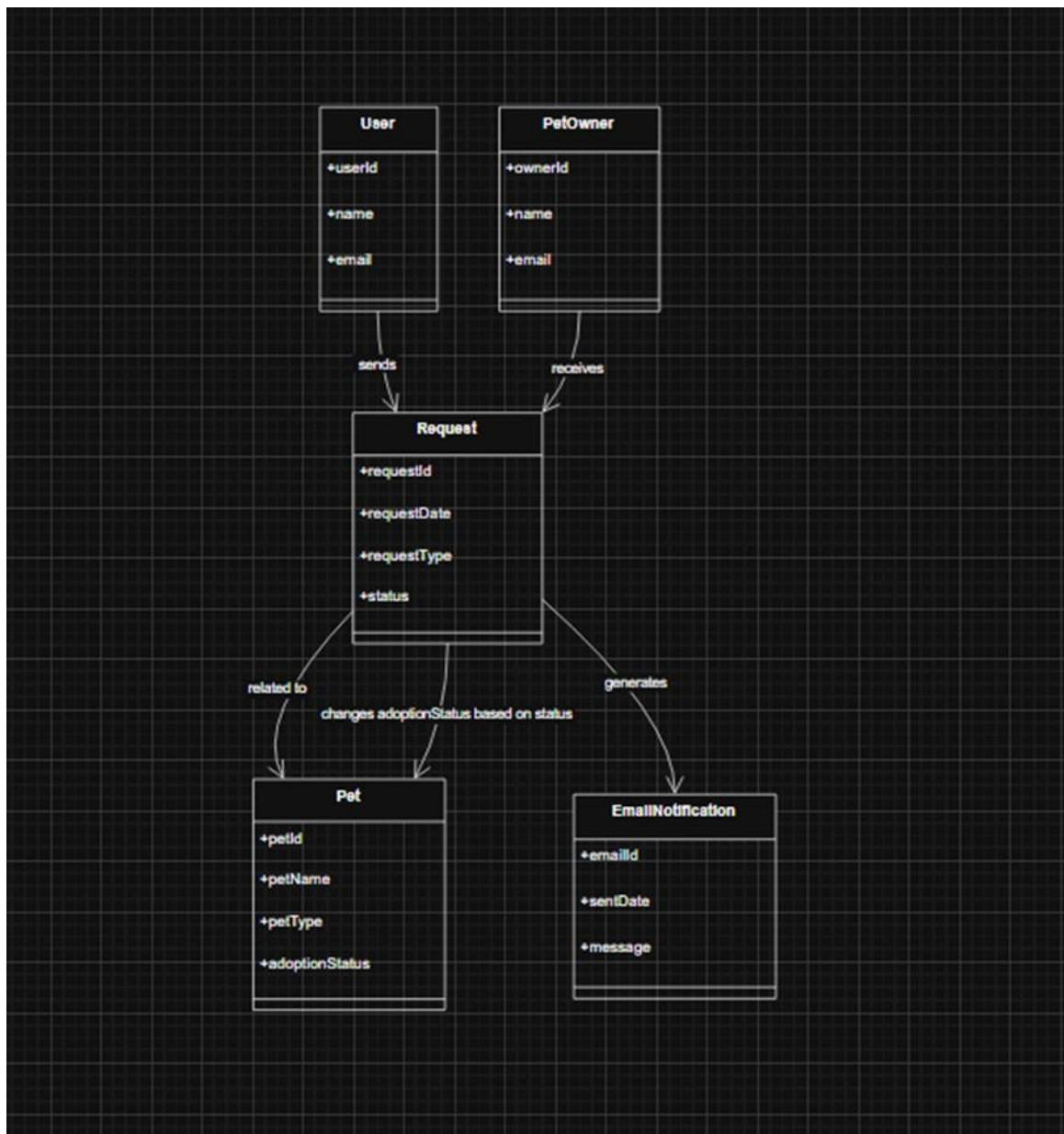


Figure XXIV: Adoption and Buying Request – Class Diagram

This diagram is defining Request classes and showing attributes like requestID, petID, userID, status, and date. It is displaying methods that are processing these requests.

5.3 Erd Diagram

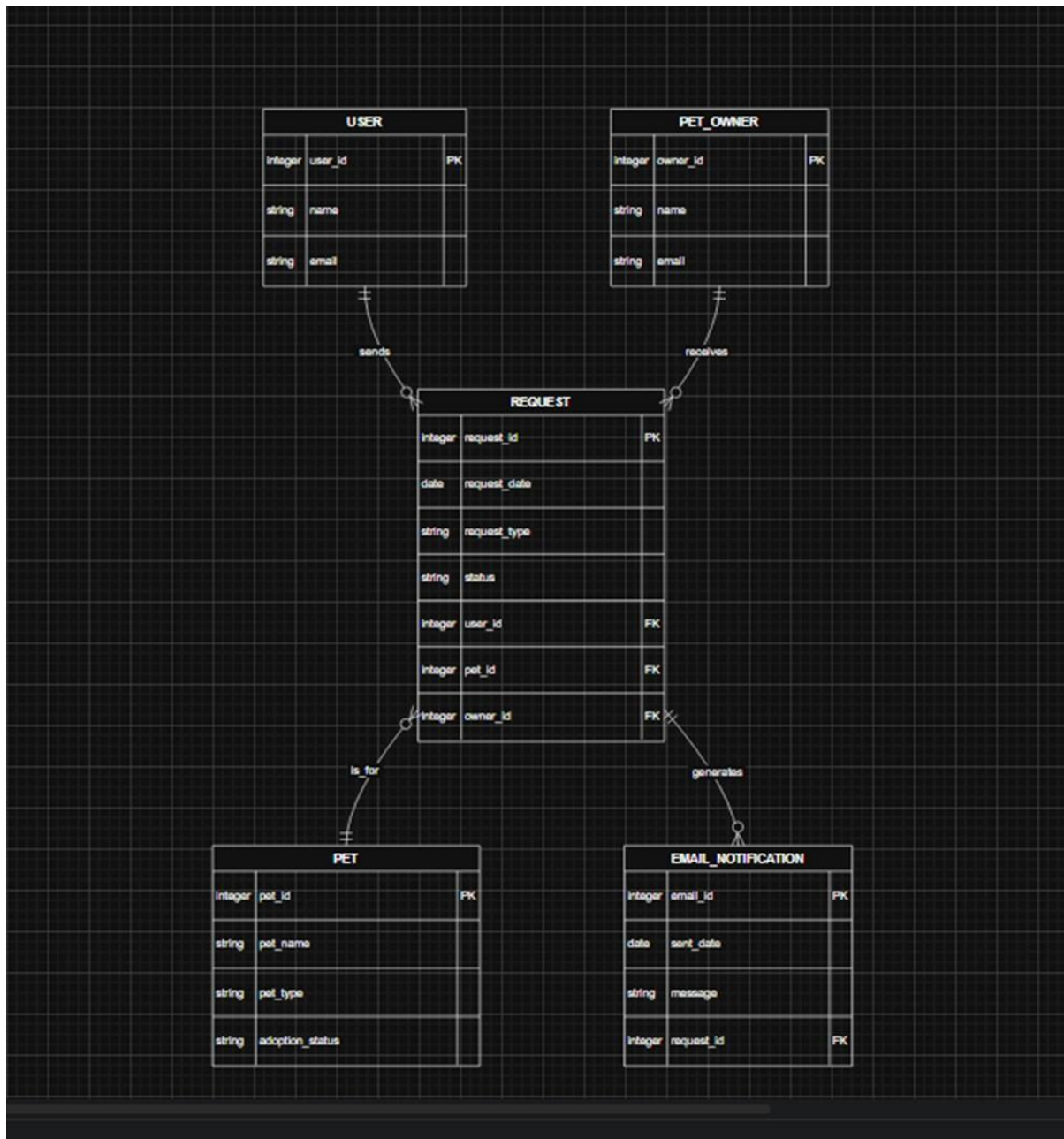


Figure XXV: Adoption and Buying Request – ERD Diagram

This diagram is showing how requests are relating to pets, users, and owners. It is illustrating the complete data structure for adoption and purchase processes.

5.4 Sequence Diagram

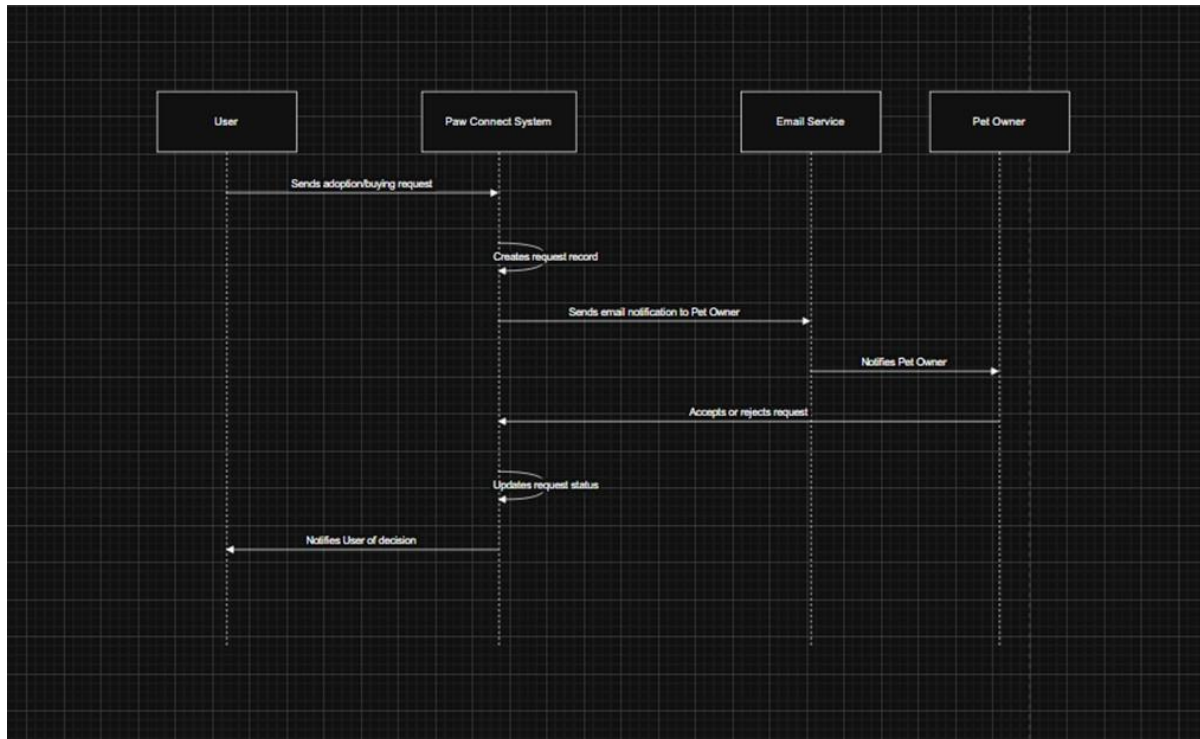


Figure XXVI: Adoption and Buying Request – Sequence Diagram

This diagram is demonstrating how messages are flowing during request submission, processing, approval, and notification stages.

5.5 UseCase Diagram

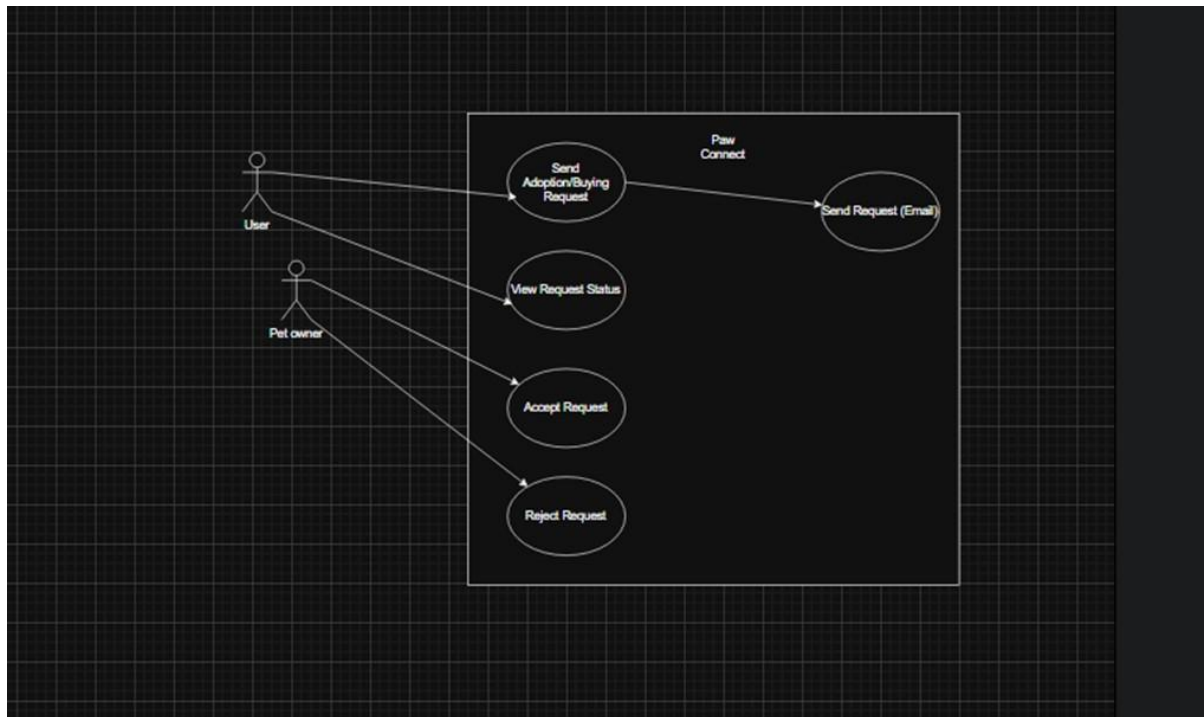


Figure XXVII: Adoption and Buying Request – Use Case Diagram

This diagram is displaying what users can do like submitting requests and tracking status, and what owners can do like approving or rejecting requests.

6.2.6.Appointment Booking

6.1 Activity Diagram

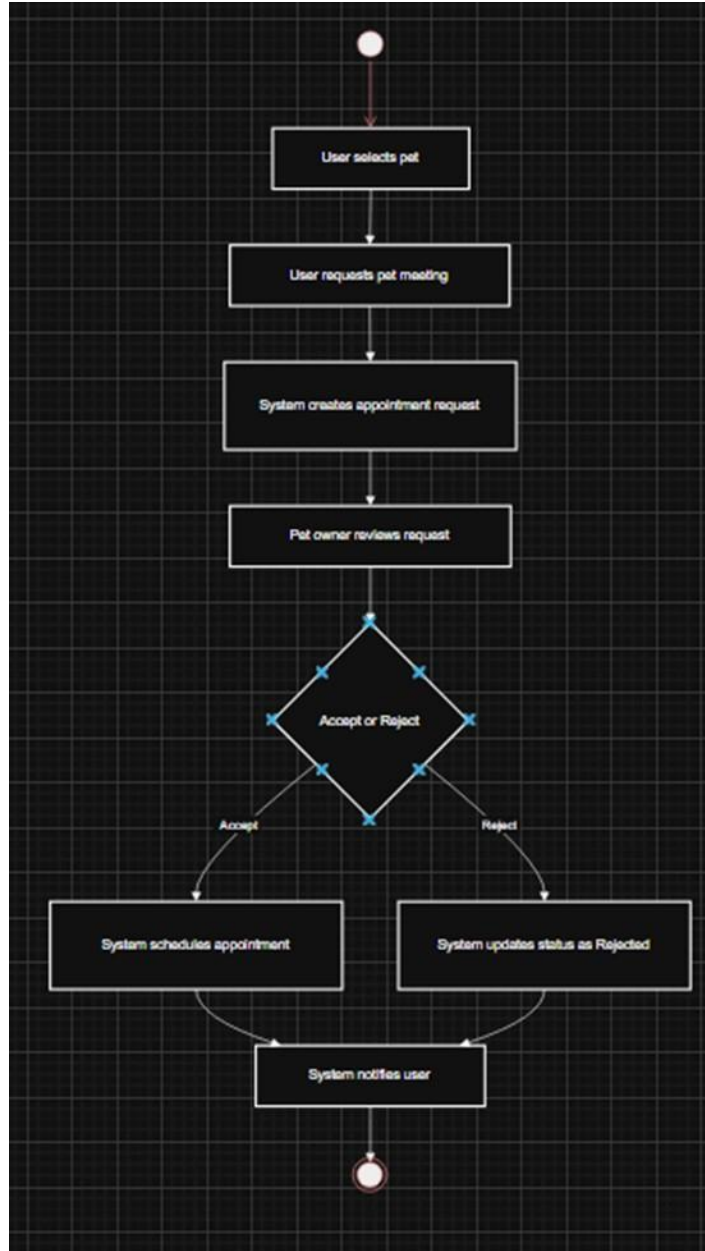


Figure 27: Appointment Booking – Activity Diagram

This diagram is illustrating how the booking process is working, showing availability checking, appointment confirmation, and notification sending.

6.2 Class Diagram

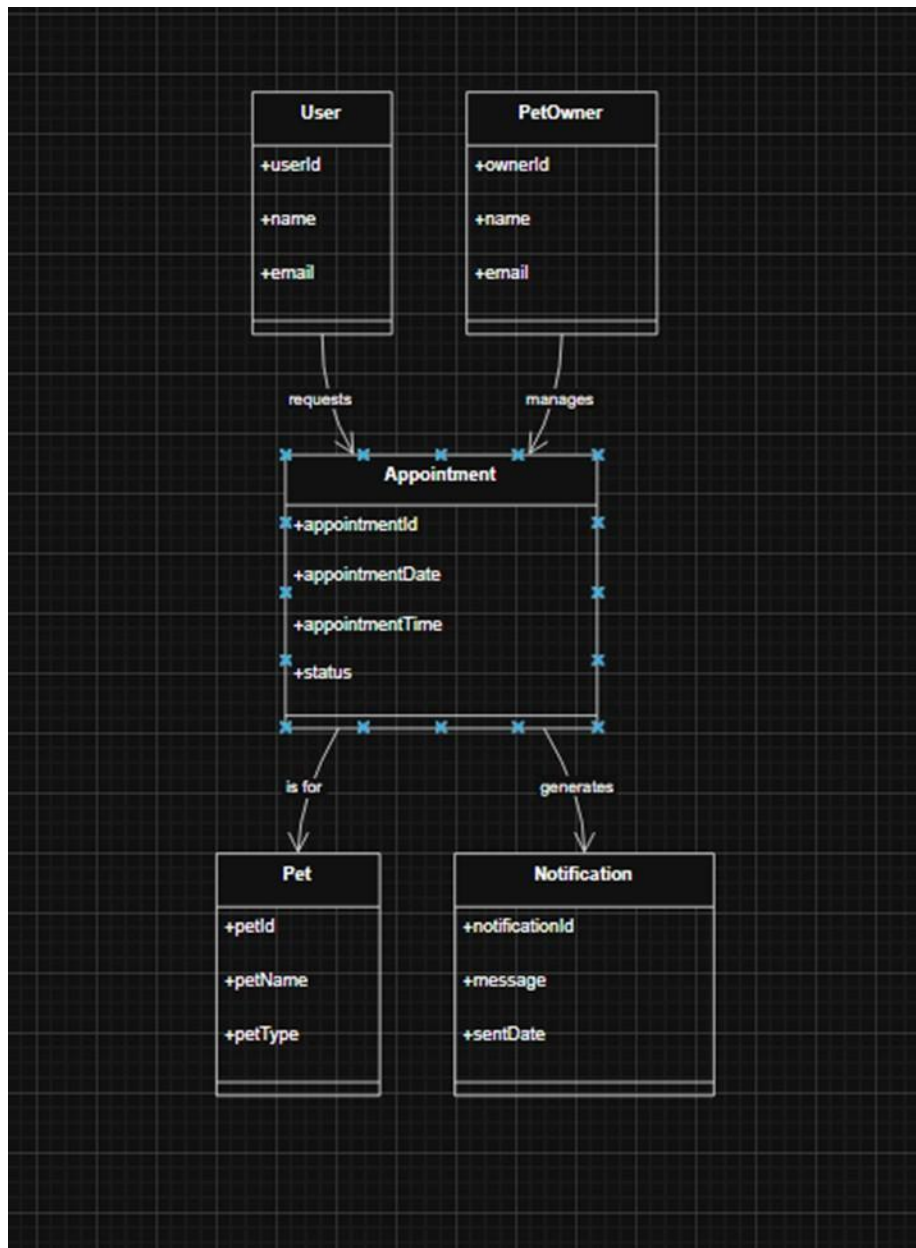


Figure XXVIII: Appointment Booking – Class Diagram

This diagram is representing the Appointment class and showing attributes like appointmentID, date, time, purpose, and status. It is displaying methods that are handling bookings.

6.3 Erd Diagram

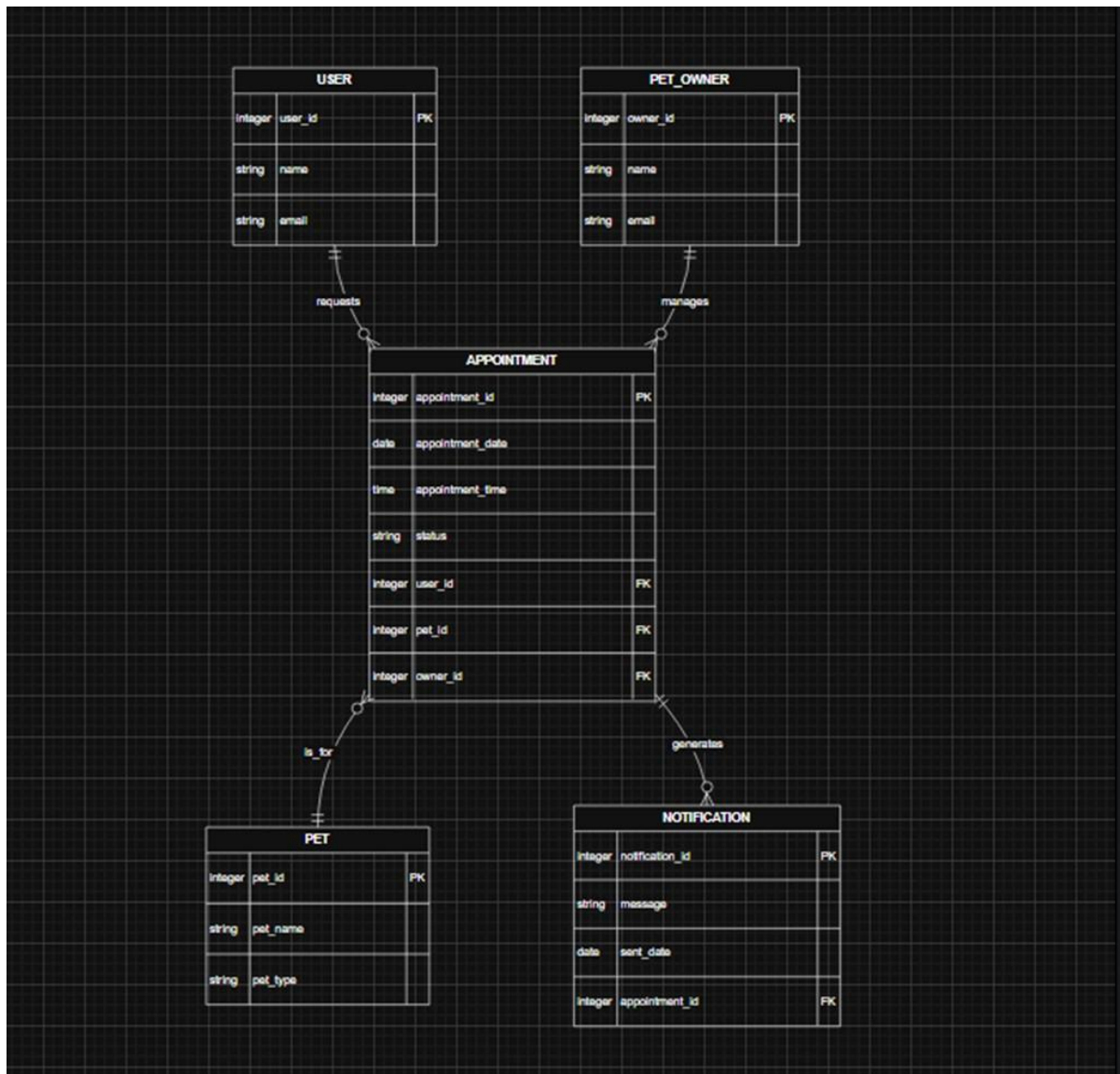


Figure XXIX: Appointment Booking – ERD Diagram

This diagram is showing the Appointment entity and illustrating how it is connecting to users, pets, and possibly veterinary services.

6.4 Sequence Diagram

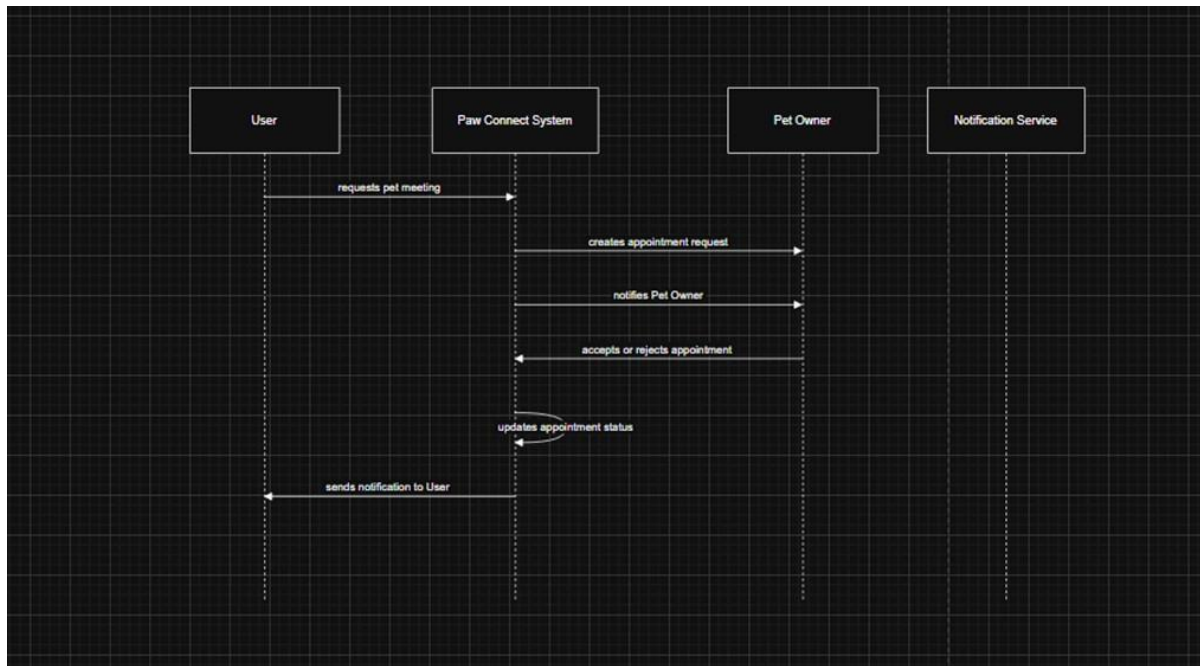


Figure XXX: Appointment Booking – Sequence Diagram

This diagram is depicting how interactions are flowing during appointment creation, modification, and cancellation processes.

6.5 UseCase Diagram

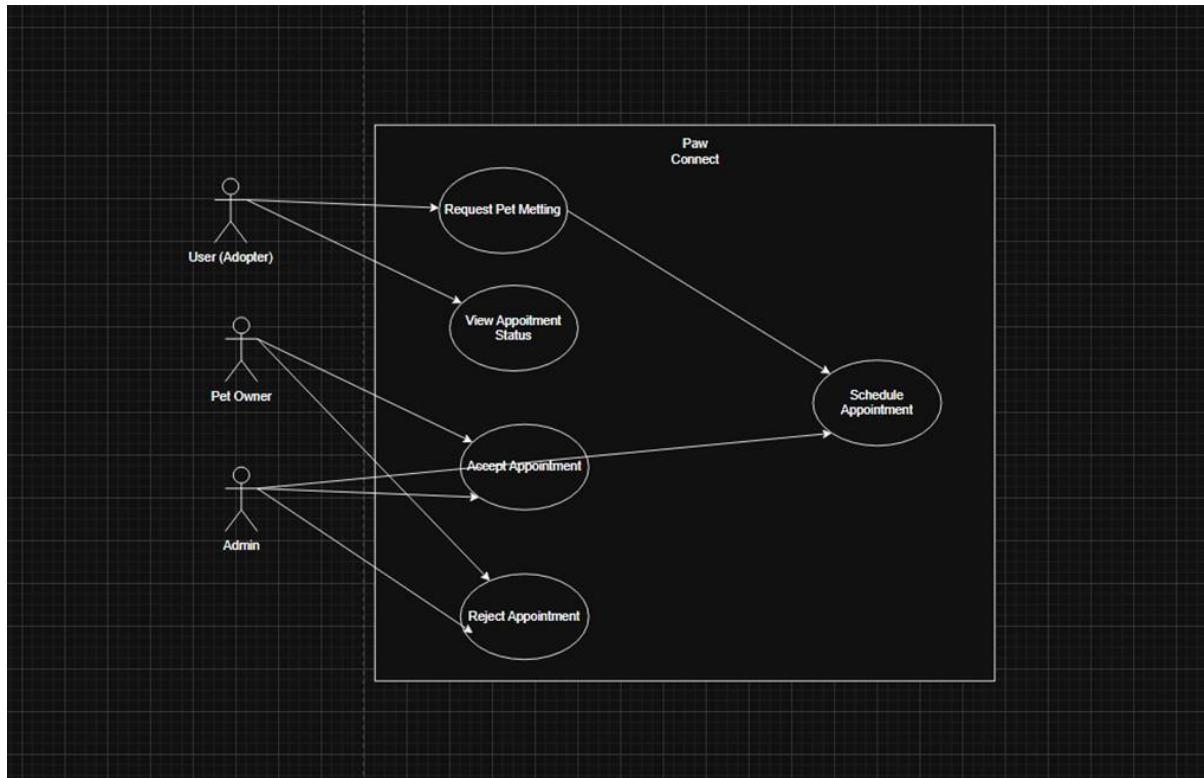


Figure XXXI: Appointment Booking – Use Case Diagram

This diagram is showing booking functionalities that users can access, displaying actions like scheduling appointments, viewing bookings, modifying schedules, and canceling appointments.

6.7 Wireframe and UI-UX design

6.7.1 System Design

The design of the PawConnect system was created to ensure a user-friendly, responsive, and visually appealing web application. The system interface was designed using Figma, which helped in planning the layout, color scheme, navigation structure, and overall appearance of the application before development. Designing the system in Figma made it easier to visualize the application structure and improve the user experience.

The design process focused on creating simple navigation, organized page layouts, and interactive user interfaces for features such as user registration, pet listings, adoption

requests, and chat functionality. The design also followed responsive design principles to ensure compatibility across desktop and mobile devices.

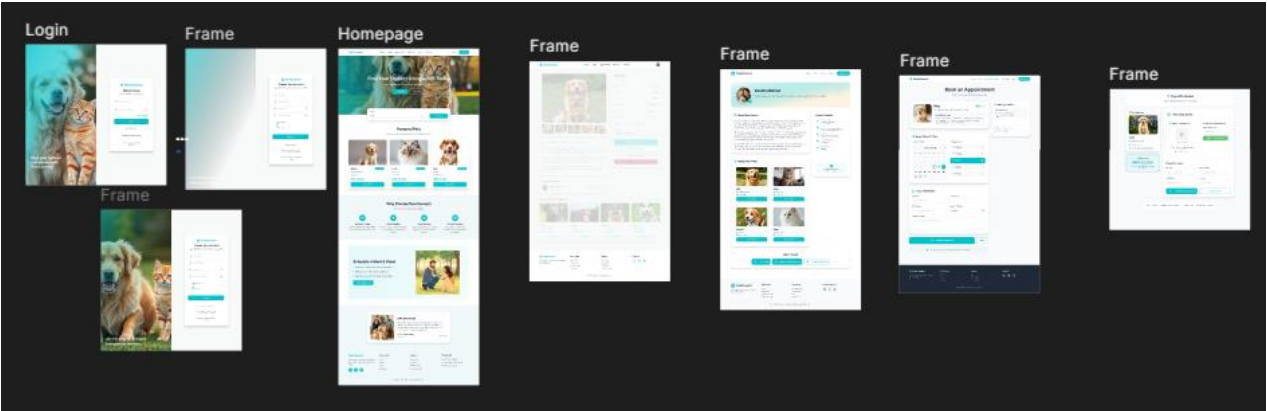


Figure XXXII: System Design – Figma UI Layout

6.7.2 Wireframe

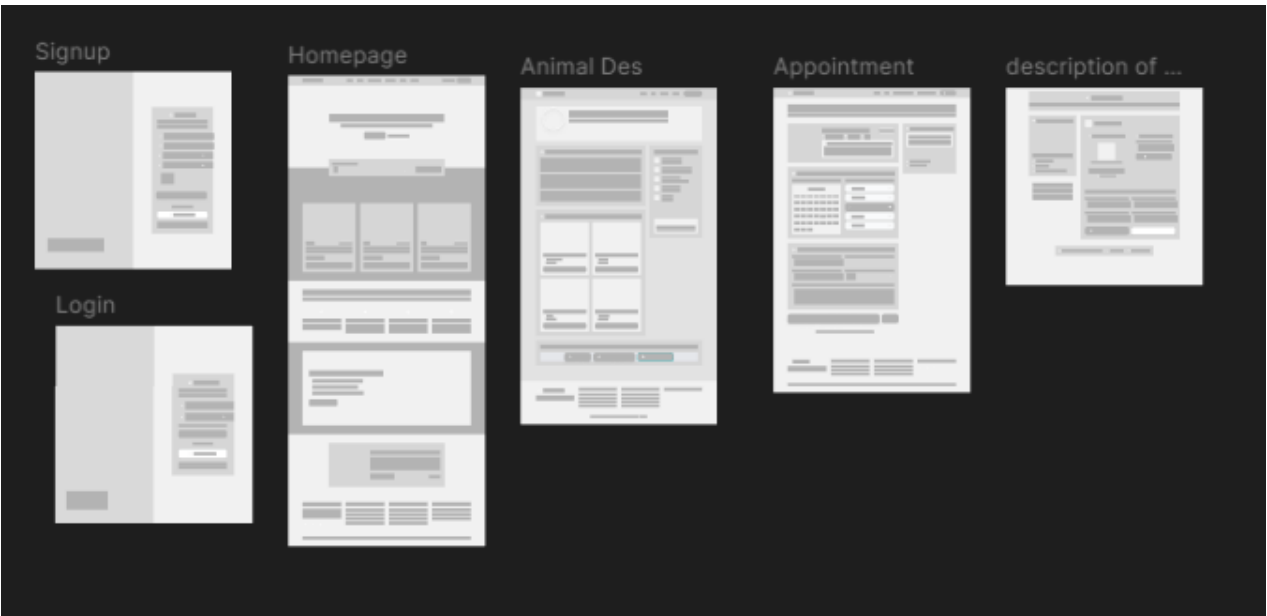


Figure XXXIII: System Wireframe – PawConnect Interface

The wireframe of the PawConnect system was developed during the initial stage of the project to provide a basic visual representation of the system structure and functionality. The wireframe included important pages such as the home page, login page, pet listing page, user dashboard, adoption request section, and chat interface.

The wireframe helped identify the placement of buttons, menus, forms, and navigation components before the actual implementation process. It also improved project planning by providing a clear understanding of how users would interact with the system. Using wireframes reduced design-related issues during development and ensured better organization of the application interface.

6.8 Testing

Test Case ID	Feature	Test Scenario	Test Steps	Expected Result	Status
TC-01	Signup	Signup with valid details	Enter name, email, password → Submit	Verify your email	Pass
TC-02	Login	Invalid email format	Enter wrong email format	Please include @ in the email address	Pass
TC-03	Forget password	Reset password with valid email	Enter registered email and submit	Password reset message sent	Pass

TC-04	Pet Adoption	User wants to adopt a pet	1. Open adoption page 2. Select a pet 3. Send request to meet pet	If user selects Yes → Adoption process continues successfully If user selects No → Adoption is cancelled	Pass
TC-05	Homepage	Page loading	Open homepage	Homepage loads correctly	Pass
TC-06	Chat	Chat functionality	Chat if u feel stuck	Open chat and send messages	Pass
TC-07	Review	Add review	Go to review and add the review	Review added	Pass
TC-08	Search	Pet type: Labrador Retriever	Go to search	To get search pet	Pass

6.8.1 Proofs of Test cases

TC-01 Verify your email



Verify Your Email

We've sent a verification email to:

pabop56803@nriza.com

 Next Steps:

Check your email inbox

Click the verification link in the email

You'll be automatically logged in

[Resend Verification Email](#)

[Back to Login](#)

Figure XXXIV TC-01: Email Verification Screen

TC-02 Please include @ in the email address



Welcome Back

Login to continue adopting pets

 Please include an '@' in the email address. 'jdjdjdmmdskkd' is missing an '@'.

[Forgot Password?](#)

Login

OR Continue With

 Continue with Google

Don't have an account? [Sign up](#)

Figure XXXV TC-02 Please include @ in the email address

TC-03 Password reset message sent



Check Your Email

We've sent password reset instructions to
bibhutikoirala702@gmail.com

Didn't receive the email? Check your spam folder or check
the backend console for the reset link.

[Try Another Email](#)

[← Back to Login](#)

Figure XXXVI TC-03 Password reset message sent

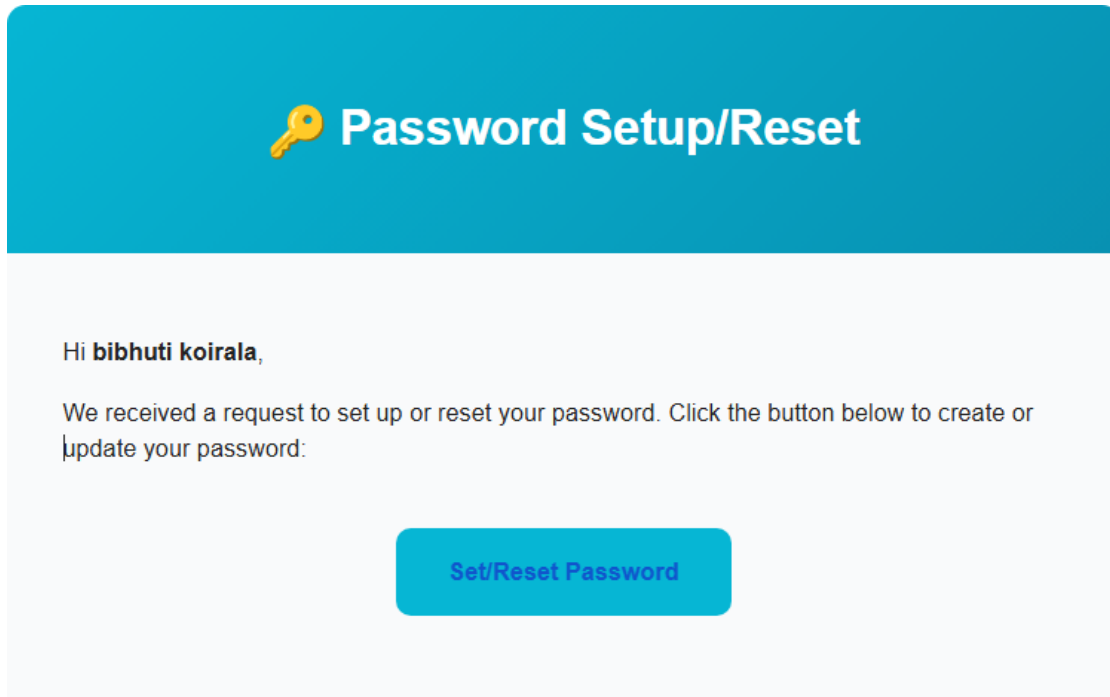


Figure XXXVII TC-04a: Adoption Request – Confirmation Prompt

TC-04 If user selects Yes → Adoption process continues successfully

If user selects No → Adoption is cancelled

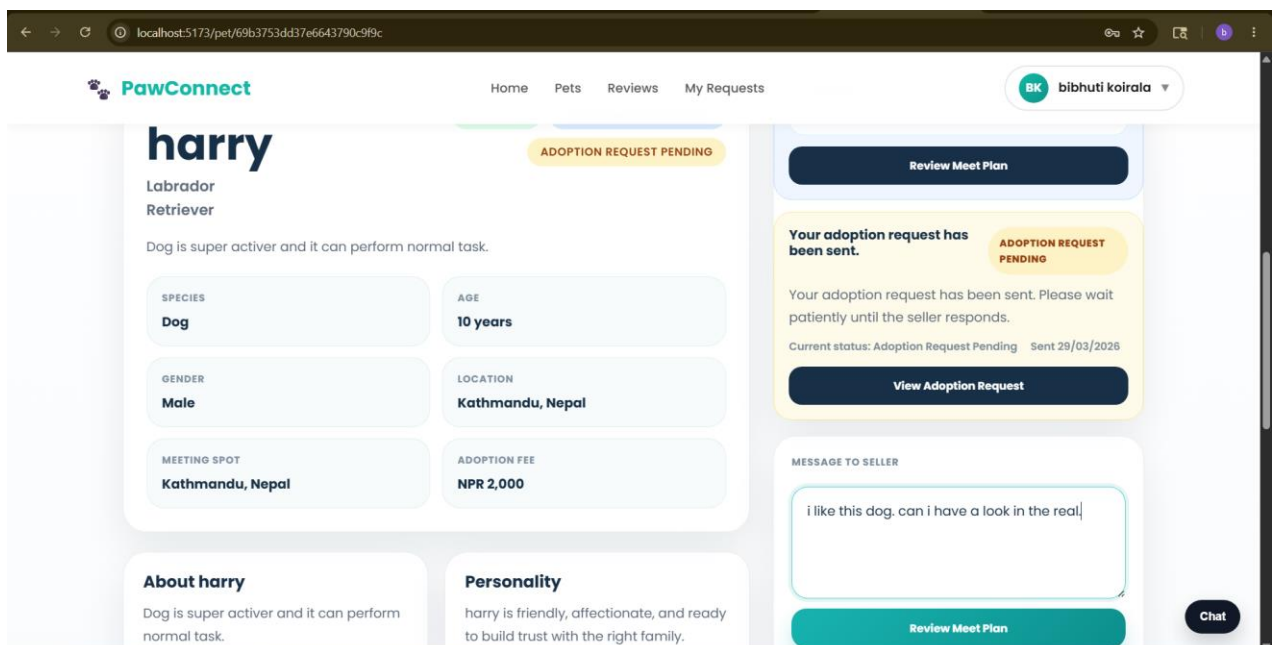


Figure XXXVIII TC-04 If user selects Yes → Adoption process continues

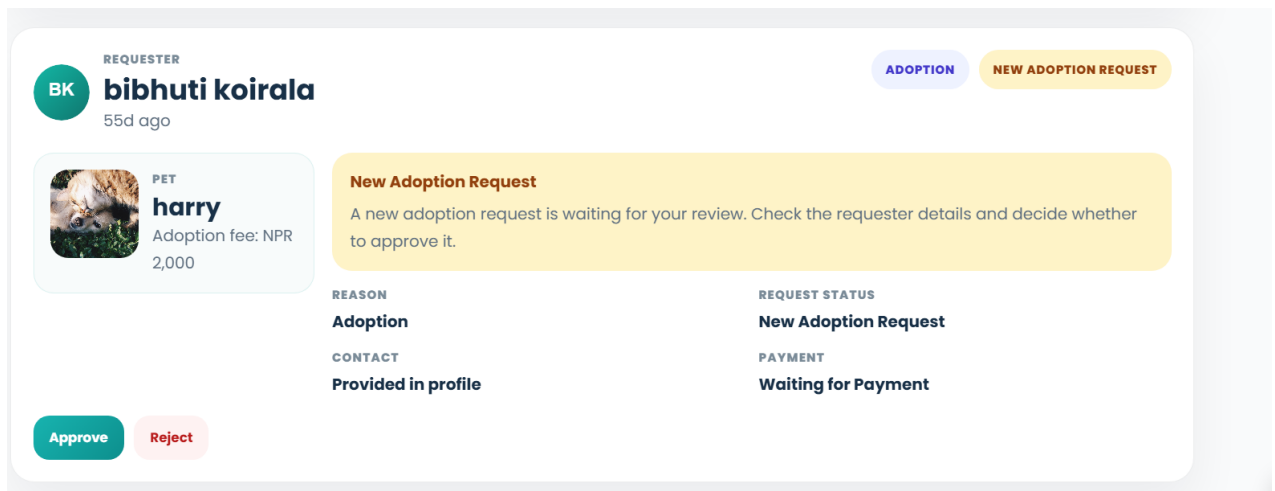


Figure XXXIX TC-04a: Approve or reject

TC-05 Homepage loads correctly

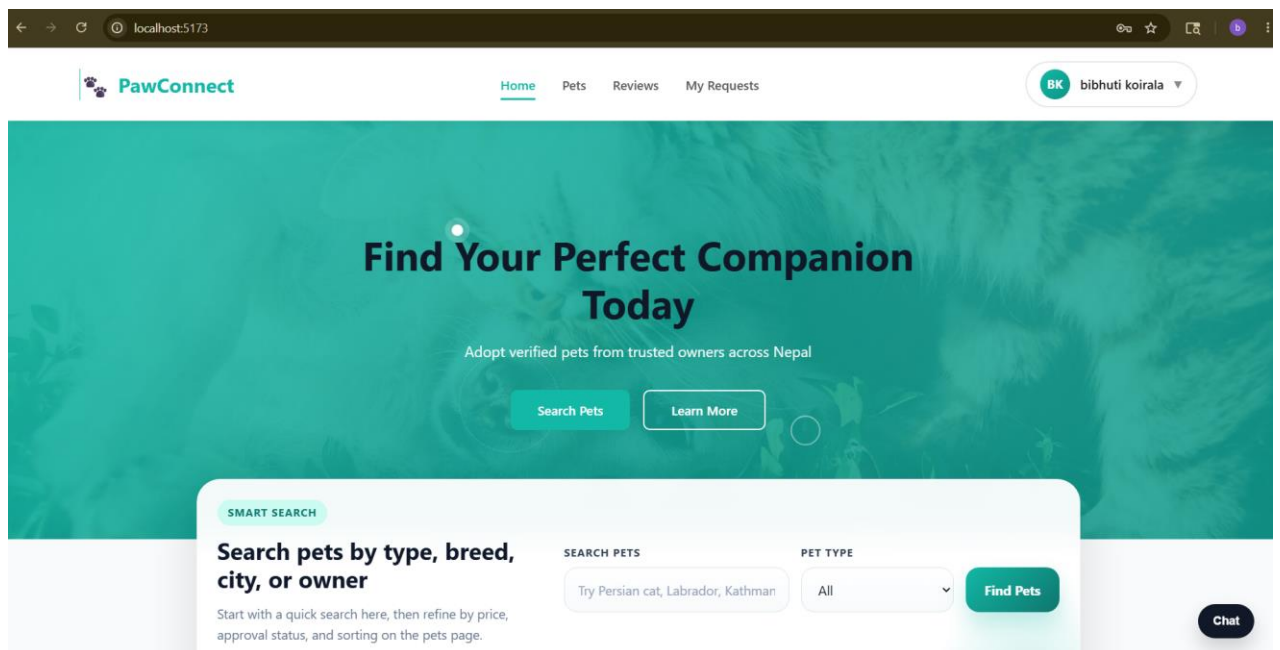


Figure XL TC-05: Homepage Loaded – Full View

TC-06 Open chat and send messages

PawConnect Assistant
Ask about adoptions or payments

Clear X

Hi! I'm PawConnect Assistant. I can help with adoptions, appointments, and payments. What do you need?

adoption

To adopt, browse pets, open a pet profile, and click Adopt. I can guide you through the steps.

Type your message...

Send

Figure XLI TC-06 Open chat and send messages

TC-07 Review added

The screenshot shows the 'Reviews' section of the PawConnect application. On the left, there is a list of reviews for both dogs and cats. Each review includes the reviewer's name, location, date, and a link to the full review. The reviews are: Prerana Bista (Coco, Kathmandu, Nov 2025), Anisha Karki (Snowy, Bhaktapur, Oct 2025), Nabin Gurung (Rocky, Pokhara, Sep 2025), and Samiksha Shrestha (Luna, Lalitpur, Aug 2025). On the right, there is a form to add a new review. The form includes fields for 'REVIEW TITLE', 'YOUR FULL REVIEW', and 'LOCATION'. The 'LOCATION' field is pre-filled with 'Kathmandu, Nepal'. There is a 'Publish review' button at the bottom of the form. A 'Chat' button is visible in the bottom right corner.

PawConnect Home Pets Reviews My Requests BK bibhuti koirala

quickly. The owner and platform were both easy to work with.

Prerana Bista
Coco · Kathmandu, Nepal
Nov 2025
[Full review](#)

Anisha Karki
Snowy · Bhaktapur, Nepal
Oct 2025
[Full review](#)

DOG ★★★★★

Confident before adoption day

The photos, pricing, and approval process were clear, so we felt confident before adoption day.

Nabin Gurung
Rocky · Pokhara, Nepal
Sep 2025
[Full review](#)

CAT ★★★★★

Smooth and trustworthy adoption

Luna was exactly as described, and the owner answered every question before we met in person.

Samiksha Shrestha
Luna · Lalitpur, Nepal
Aug 2025
[Full review](#)

REVIEW TITLE
Smooth and trustworthy adoption

YOUR FULL REVIEW
Tell the full story of your adoption journey...

LOCATION
Kathmandu, Nepal

Your Review Center settings currently allow public reviews: **Yes**

Publish review

Chat

Figure XLII TC-07: Review Added Successfully

TC-08 Search pet

The screenshot shows the 'Pets' section of the PawConnect application. The search bar is pre-filled with 'Labrador Retriever'. The search results show a list of pets, with the first one being 'harry', a Labrador Retriever, available for NPR 2,000. The search filters include 'All statuses', 'Sort by: Newest first', 'Min Price: 0', and 'Max Price: 566600'. A 'Reset' button is also present. The 'Price range' is displayed as 'NPR 2,000 - NPR 566,600'. A 'Chat' button is visible in the bottom right corner.

PawConnect Home Pets Reviews Adoption Requests BK Bibhuti Koirala

All pets Dogs Cats Birds Rabbits Other pets

SEARCH **STATUS** **SORT BY** **MIN PRICE** **MAX PRICE** **Reset**

Labrador Retriever All statuses Newest first 0 566600

Showing 1 of 11 pets Price range: NPR 2,000 - NPR 566,600

harry **NPR 2,000**
Labrador Retriever

Chat

Figure XLIII TC-08: Search – Pet Type Filter Result

7. Conclusion

7.1 Review of Aims, Objectives, and Academic Questions

The main aim of the PawConnect project was to develop a web-based platform that simplifies the pet adoption and purchasing process by connecting pet owners, shelters, and users interested in pets. The system was designed to improve communication, organize pet information, and provide an efficient adoption workflow.

The objectives of the project were successfully achieved by implementing features such as pet listing management, search and filtering functionality, adoption and purchase request handling, appointment scheduling, and chat communication.

The academic question of the project focused on understanding how digital information systems can support real-world adoption processes. The development of PawConnect demonstrated how software engineering concepts, system design methodologies, and web technologies can be used to solve practical problems.

7.2 Findings and Conclusions

From the development and implementation of the PawConnect system, several conclusions can be drawn:

- Web-based platforms can improve the efficiency and accessibility of pet adoption services.
- Structured digital communication helps reduce misunderstandings between users and pet owners.

- Search and recommendation features help users find suitable pets faster.
- Integrating multiple adoption-related functions into one platform improves user experience.
- Machine learning-based recommendation concepts can be applied to improve future system intelligence.

Overall, the PawConnect system demonstrates how information technology can be used to modernize the pet adoption process.

8. Critical Evaluation of the Project

8.1 Final Report Reflection

This project report presents the design and development of the PawConnect system. The report includes system analysis, literature review, design artefacts, implementation approaches, and testing results.

The report demonstrates good integration of theoretical knowledge and practical system development. The use of diagrams, structured subsystem explanation, and testing cases helped in presenting the project clearly.

8.2 Findings and Development Process

During the project development process, several important findings were identified.

The modular system design helped simplify development by dividing the system into smaller functional components.

Using React for frontend development improved interface responsiveness and user interaction.

ASP.NET Core provided a stable and secure backend environment for handling system operations.

Testing during development helped detect and fix errors early, improving overall system quality.

8.3 System Evaluation

The PawConnect system successfully provides core functionalities such as user authentication, pet management, search filtering, request handling, appointment booking, and chat communication, add review.

The system is user-friendly and supports organized adoption workflow. However, some advanced features such as real-time messaging optimization, payment integration, and veterinary verification are included.

8.4 Planning, Management, and Quality of Sources

The project was managed using an incremental development approach that allowed continuous improvement during the implementation process. Different modules were developed and tested step by step to ensure smooth progress and better system functionality.

Research sources were selected from reliable online platforms, official documentation, and academic studies related to pet adoption systems and web application development. The quality of the system was maintained through proper coding practices, modular design, and regular testing procedures.

8.5 Self-Reflection

Developing this project helped improve my understanding of full-stack web development and software engineering principles.

Through this project, I learned about system design, database relationships, API development, and user interface implementation.

The project also improved my problem-solving skills, technical documentation ability, and knowledge of machine learning-based recommendation concepts.

Overall, this project contributed positively to my academic and professional development.

9. Evidence of Project Management

9.1 Log Sheet

https://drive.google.com/file/d/1D_-ujVQn-1TC0UZX8zKePZW6-Fax4klD/view?usp=sharing

A weekly logsheet to record the progress of my work. In the logsheet, I documented the tasks I completed during the previous week and outlined the tasks planned for the following week. The logsheet was then reviewed and signed by my teacher accordingly.

Declaration Sheet

Declaration Sheet

Award Title: BSc(Hons) Computer Science
(Award Title for your project, if in doubt refer to your course/Module Registration)

Declaration Sheet
(Presented in partial fulfillment of the assessment requirements for the above award.)

This work or any part thereof has not previously been presented in any form to the University or to any other institutional body whether for assessment or for other purposes. Save for any express acknowledgements, references and/or bibliographies cited in the work. I confirm that the intellectual content of the work is the result of my own efforts and of no other person.

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Student Name: *Bibhuti Koirala*

Student ID Number: *2417754*

Signature: *Bibhuti* Date: *22 May 2026*
(Must include the unedited statement above. Sign and date)

Please use an electronic signature (scan and insert)

Figure 44 Declaration Sheet

10. Appendices

10.1 User Manual

1. Register and Login

- Click the Register button and enter the required details.
- Submit the form to create an account.
- For login, enter your email and password and click Sign In.

2. Search Pets

- Navigate to the Search/Pets section.
- Use the search bar or filters to find pets.
- Click on a pet listing to view detailed information.

3. Send Adoption Requests

- Open the pet profile page.
- Click Send Adoption Request.
- Fill in the required information and submit the request. • Wait for the owner's approval.

4. Chat System

- Go to the Chat section.
- Text were if u feel stuck
- Click Send.
- It respond u back.

5. Book Appointments

- Go to the Appointments section.
- Select appointment type, date, and time.
- Fill in the required details and click Book.
- Confirmation will be sent through notification or email.

6. Add Reviews

- Navigate to the **Review** section.
- Write your feedback and rating about the service or pet adoption experience.
- Submit the review successfully.

10.2 System Configuration

1. Software Requirements

- Web Browser: Google Chrome, Microsoft Edge, or Mozilla Firefox
- Code Editor: Visual Studio Code or Visual Studio
- Runtime Environment: Node.js for React development

- Backend Framework: ASP.NET Core (.NET)

2. Framework Versions

- Frontend: React.js
- Backend: ASP.NET Core

3.Database Configuration

The system uses MongoDB as a NoSQL database to store user data, pet listings, adoption requests, appointment details, and chat messages. MongoDB was chosen because it supports flexible and efficient data storage for web applications.

4.Server Setup Details

The backend is developed using ASP.NET Core Web API, while the frontend runs on the React development server or production hosting server. The system communicates through secure and authenticated API endpoints.

11. Future Enhancements

- Development of a mobile application for Android and iOS platforms.
- Integration of AI-based pet recommendation systems.
- Real-time notifications and messaging improvements.
- Veterinary verification and pet medical record integration.
- Advanced analytics for adoption trends and user behavior.
- GPS-based search for nearby pets and shelters.

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[MDN Web Docs](#)
- [21] [Petfinder](#)
- [22] [Adopt a Pet](#)
- [23] [Humane Society International](#)
- [24] [ASPCA Adoption Center](#)

[25] [Rescue Me](#)